Advanced bio-adaptive hydrogels in spinal cord injury repair: from microenvironment remodeling to functional recovery

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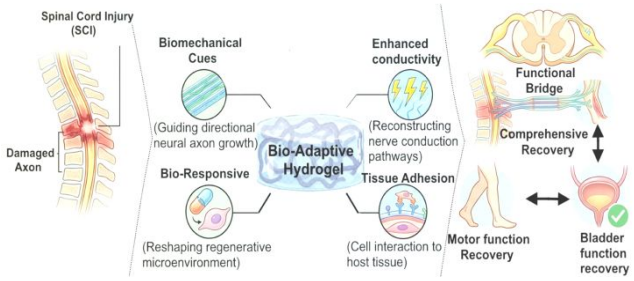
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Abstract

Spinal cord injury (SCI) presents a huge clinical challenge in neurological repair and functional reconstruction. The advancement of regenerative medicine and neuroengineering technologies has positioned bio-adaptive hydrogels as an innovative therapeutic strategy for SCI treatment. Their controllable (bio)physicochemical properties, excellent biocompatibility, and ability to mimic the extracellular matrix (ECM) of the spinal cord allow them to effectively reshape the regenerative microenvironment, guiding the directional growth of neural axons and reconstructing nerve conduction pathways. Unlike previous reviews that focused on material synthesis or single-tissue repair, this article systematically elaborates on the multidimensional design strategies (e.g., biomechanical cues, enhanced conductivity, dynamic biological responses, and interfacial tissue adhesion) of bio-adaptive hydrogels for active regulation and rehabilitation of SCI. By summarizing recent research progress, we critically analyze the often-overlooked intrinsic connection between motor and bladder function repair, laying a theoretical foundation for achieving comprehensive functional recovery after SCI. Finally, recognizing the need for more intelligent and responsive therapeutic platforms, this article explores the emerging applications of artificial intelligence (AI) in the rational design and preparation of hydrogels, outlining a direction for the development of next-generation, AI-assisted smart hydrogels.

Keywords: spinal cord injury, tissue microenvironment, bio-adaptive hydrogel, functional recovery

Graphical Abstract



1. Introduction

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Spinal cord injury (SCI) is a serious central nervous system(CNS) disorder that causes dysfunction of the motor, sensory, and autonomic nervous systems due to damage to the spinal cord [1]. Data from the 2019 Global Burden of Disease Study (GBD 2019) show that the number of current patients with SCI worldwide has exceeded 20 million, and the annual number of new cases has reached approximately 900,000. Most cases are caused by external trauma, including falls (477,000 cases, the primary cause), road traffic accidents (230,000 cases, the second leading cause), and violent incidents [2]. Surgical decompression and anti-inflammatory treatment during the acute phase have significantly reduced early mortality. However, the permanent functional loss associated with SCI continues to leave patients confronting the challenge of lifelong disability. Additionally, the high incidence of long-term complications, including neurogenic bladder, chronic intractable pain, deep vein thrombosis, and pressure ulcers, severely impacts patients' quality of life and imposes a substantial economic burden on their families [3-5]. Consequently, promoting neurological recovery and improving the prognosis following SCI represent critical issues that warrant urgent attention in contemporary clinical research.

Pathologically, initial mechanical trauma to the spinal cord, also known as primary injury, directly disrupts spinal cord structures, causing neuronal necrosis, axonal disruption, and vascular rupture, leading to irreversible damage [6]. The subsequent secondary injury is the key factor in disease deterioration. Inflammatory reaction, oxidative stress, and excitatory amino acid toxicity create a harsh microenvironment that further triggers a cascade of events, including apoptosis, abnormal astrocyte activation, and glial scar formation [7-9]. Glial scar and deposited chemical inhibitory molecules (such as chondroitin sulfate proteoglycans (CSPGs)) jointly constitute a physical barrier, that limits the regeneration of endogenous axons and the reconstruction of neural circuits [10, 11].

Secondary injury following SCI is a dynamically evolving pathological cascade that can be temporally delineated into three phases: acute, subacute, and chronic. Upon occurrence of the primary mechanical insult, the acute secondary injury phase is immediately initiated and typically lasts up to 48 hours post-injury. Its hallmark pathological features include disruption of vascular integrity, ionic homeostasis imbalance, glutamate-mediated excitotoxicity, massive generation of reactive oxygen species (ROS), intracellular calcium overload, lipid peroxidation, focal inflammatory

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bursts, tissue edema, and widespread cell necrosis [8, 12, 13]. If the acute injury cascade is not effectively controlled, it progresses to the subacute secondary injury phase, generally spanning from day 2 to week 2 post-injury, which is characterized mainly by neuronal apoptosis, extensive demyelination of axons, Wallerian degeneration of distal axonal segments, initiation of local axonal remodeling attempts, and proliferation of reactive glial cells with the beginning of glial scar formation [8, 12]. The pathological events of the subacute phase ultimately drive the lesion into the chronic secondary injury phase, which begins beyond 2 weeks after injury and typically persists for months to years. Its core features are the formation of a cystic cavity at the lesion center, the retraction of surviving axons, and the full maturation and consolidation of the glial scar, creating physical and biochemical barriers that impede neural regeneration [8, 12, 14].

Traditional treatments for SCI have numerous limitations and have yet to show satisfactory therapeutic effects. Although surgical procedures can maintain spinal stability and achieve local decompression, there is still controversy over the optimal timing of the operation[15, 16]. Drug therapy is limited by the barrier effect and the non-targeted distribution of the blood-spinal cord barrier (BSCB), making it difficult to achieve effective therapeutic concentrations at the lesion site [17, 18]. Rehabilitation training and functional electrical stimulation can help maintain joint range of motion and improve residual motor functions, but their precise mechanisms remain incompletely understood. Additionally, functional gait recovery has not been clearly observed in animal models of chronic SCI [19]. Furthermore, cell transplantation therapies (such as neural stem cells (NSCs) and bone marrow mesenchymal stem cells (BMSCs)) have relatively low survival rates, differentiation, and directional migration efficiency in pathological microenvironments, such as inflammatory responses and glial scars, resulting in poor therapeutic effects [19, 20]. Finding a novel therapeutic strategy to overcome the aforementioned constraints and enhance neural regeneration and functional recovery following SCI is, therefore, highly important in biomedicine and tissue engineering.

In recent years, biomaterials have emerged as a focal point of research in this field. Among these, hydrogels have distinguished themselves for their remarkable properties, including swelling and shrinking capabilities, viscoelasticity, biocompatibility, degradability, and adjustable mechanical strength, characteristics that offer significant

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advances for SCI repair [21-24]. As a 3D cross-linked hydrophilic polymer network, a hydrogel can effectively inhibit excessive glial scar proliferation by filling the injury cavity [25, 26]. However, in most prior studies, hydrogels have primarily functioned as simple bridging materials or as delivery vehicles for drugs, cells, and neurotrophic factors. While they serve as a structural scaffold, these hydrogels often lack biological activity and are ineffective at mitigating secondary injuries or facilitating active interactions with spinal cord tissue [27-29]. This limitation has driven the field towards the development of "bio-adaptive" hydrogels designed not merely as passive fillers, but as dynamic platforms that actively participate in tissue repair by modulating cellular behavior and the injury microenvironment.

Bio-adaptive hydrogels are composite systems composed of active components and inert matrices. Their core advantage lies in their ability to integrate bio-adaptive substances, thereby regulating the physiological processes they modulate. Through appropriate design, this type of hydrogel can create a microenvironment that supports the survival, proliferation, and directional differentiation of NSCs while also guiding axonal extension and synaptic remodeling. These characteristics make it highly applicable to neural tissue engineering. Bio-adaptive hydrogels, unlike traditional hydrogels, exhibit multifunctionality and dynamic responsiveness, actively influencing biological processes by enhancing cell proliferation and differentiation, guiding axonal growth, mimicking electrical conduction, precisely regulating the microenvironment of injured spinal cords, and promoting neural circuit reconstruction. While previous reviews have comprehensively covered material synthesis strategies or basic hydrogel applications in single-tissue repair, this review systematically organizes the multidimensional design approaches of bio-adaptive hydrogels for SCI repair and their functional recovery effects (**Figure 1**), with particular emphasis on the often-overlooked interplay between material design strategies and the integrated restoration of both motor and bladder functions. By evaluating the translational gap and exploring emerging directions, such as the application of artificial intelligence (AI) in hydrogel design, this review aims to provide valuable insights to advance the clinical treatment of SCI and achieve functional recovery.

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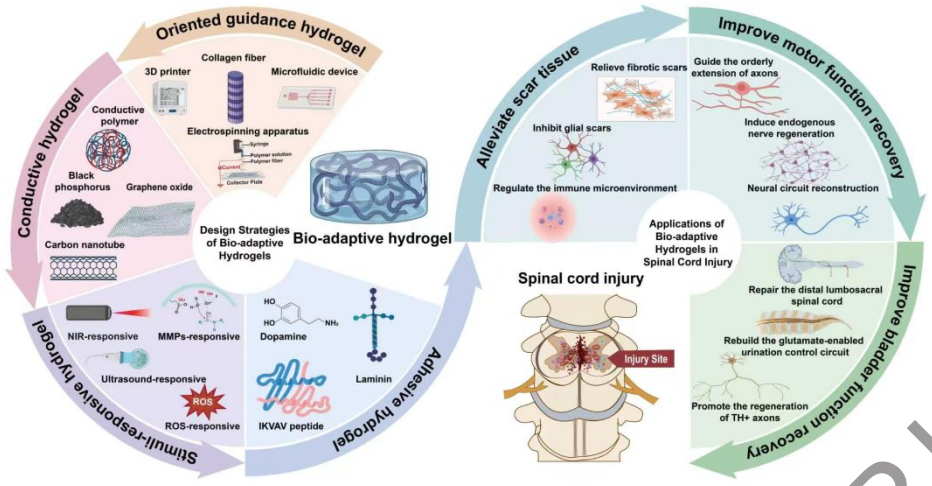


Figure 1. Design strategies and applications of bioadaptive hydrogels for repair and functional recovery following SCI. The figure was created using BioRender.com.

2. The basic characteristics of bio-adaptive hydrogels and their advantages in SCI repair

Bio-adaptive hydrogels are functional materials constructed from natural or synthetic polymers as scaffolds, whose design centers on closely mimicking the three-dimensional architecture, mechanical properties, and biochemical signaling networks of the native extracellular matrix (ECM). Through strategies such as covalent modification, physical encapsulation, or *in situ* assembly, functional building blocks including cell-adhesion sequences, targeting ligands, growth factors, and stimuli-responsive moieties can be incorporated into hydrogels to endow them with the basic characteristics of actively regulating cell fate and remodeling the microenvironment. This evolution transforms conventional hydrogels from mere structural supports into systems capable of signal modulation, dynamic response, and functional guidance, thereby markedly enhancing material–host tissue interactions [30]. Such bio-adaptive hydrogels can provide an optimal regenerative microenvironment for tissues that are difficult to repair and require complex microenvironmental regulation, such as neural tissue, spinal cord, and cartilage, and offer critical support for overcoming clinical barriers in refractory tissue repair and for advancing translational application of regenerative medicine materials from basic research to clinical use.

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Existing studies indicate that bio-adaptive hydrogels can effectively ameliorate the local microenvironment at the injury site, promote neurogenesis and axonal regeneration, and thereby facilitate spinal cord tissue repair [31]. Their tunable mechanical properties, particularly stiffness optimization, are critical for promoting neural regeneration and functional recovery after SCI [32]. Optimizing hydrogel stiffness not only reduces glial scar formation but also directly regulates NSC fate through mechanical signals, thereby providing a suitable mechanical microenvironment for axonal regeneration. For instance, hydrogels mimicking the soft microenvironment of the neonatal spinal cord (approximately 300 Pa) can induce axonal regeneration along a negative stiffness gradient by inhibiting the neuronal Piezo1/RhoA/MLC mechanical signaling pathway [33]. The injectable property of bio-adaptive hydrogels enables them to be delivered to the SCI site through minimally invasive means and formed *in situ*, filling the defect cavity and closely adhering to the edge of the nerve tissue, restoring the continuity of the spinal cord tissue, while avoiding the secondary mechanical damage that may be caused to the residual nerves during the implantation of traditional solid stents [34-36]. In addition, the self-healing ability of bio-adaptive hydrogels can adapt to the physiological activities of the spine, prevent rupture of scaffold structures, and ensure the long-term stability of the repair process [37]. With the continuous advancement of material synthesis and functionalization technologies, bio-adaptive hydrogels have evolved from early passive scaffolds that provide only physical support to intelligent repair materials that actively sense and regulate the injured microenvironment. By targeting and intervening at the key pathological links underlying microenvironmental imbalance, they achieve neural tissue regeneration and functional recovery.

Given the anatomical specificity and pathological complexity of the CNS, the design of bio-adaptive hydrogels for SCI must meet the following specific requirements: (i) the hydrogel modulus should precisely match native spinal cord tissue (gray matter ≈ 0.1~1 kPa; white matter can reach up to 4 kPa at most) [38]; (ii) while maintaining ultrasoft mechanical properties, the hydrogel should present anisotropic linear microchannels or parallel microfiber topographies to mimic white matter fiber tracts, guiding axons to span the lesion cavity and grow in an ordered fashion to reconstitute functional neural circuits [39]; (iii) the hydrogel should be integrated with conductive components to achieve electrical conductivities comparable to native spinal cord tissue

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(10^{-2}-10^{-1} $S \cdot cm^{-1}$), thereby supporting electrical signal transmission across the lesion [40]; and (iv) the hydrogel must accommodate the dual nature of CNS glial scarring and its immune dynamics: it should precisely balance the scar's physical barrier and chemical inhibitory properties by targeted degradation of restrictive constituents rather than wholesale removal of scar tissue to preserve structural integrity, and it should enable temporally controlled immunomodulation suppressing excessive inflammation during the acute phase and promoting a reparative phenotype in the subacute phase to limit secondary injury while creating an immune microenvironment conducive to axonal regeneration.

3. Design strategies for bio-adaptive hydrogels in SCI repair

3.1 Biomimetic mechanics and structural design of bio-adaptive hydrogel for directional guidance of axonal regeneration

A fundamental challenge in SCI repair is guiding regenerating axons across the lesion site in an organized manner. Following injury, the disordered growth of axons severely hinders the establishment of functional neural repair. To address this, bio-adaptive hydrogels have been engineered with biomimetic mechanical properties and topographical structures that provide directional cues for axonal growth. Studies have consistently demonstrated that topologically arranged structures can significantly induce directional axonal extension and promote precise neuronal connections [41-45]. While conventional porous hydrogels can serve as a structural scaffold for nerve axon regeneration, they lack the spatial cues necessary to guide directional nerve regeneration. To overcome this limitation, researchers have introduced oriented fiber structures into the hydrogel matrix. This topological guidance, combined with the hydrogel's biochemical cues, jointly guides the oriented extension of axons and cell migration, promoting both myelination and functional recovery [46]. In previous studies, researchers developed an aligned fibrin hydrogel (AFG) that combines a soft tissue–matched mechanical modulus with an oriented nanofibrous topology, thereby recapitulating the low elasticity and alignment characteristics of the ECM to achieve intimate integration with spinal cord tissue (**Figure 2A**). This hydrogel has been shown to exhibit favorable bioactivity, promoting the directed migration of endogenous NSCs, facilitating orderly axonal extension and vascular remodeling, and thereby supporting the organized recovery of motor function [47, 48]. The therapeutic efficacy of AFG has

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been validated in a large-animal (canine) SCI model, confirming its potential for clinical translation (**Figure 2B, C**). Following implantation, AFG markedly promoted white matter repair, whereas its effect on gray matter restoration was relatively limited [49]. Because spinal cord function depends on signal conduction within white matter and synaptic integration within gray matter, improvement of white matter regeneration alone is insufficient to achieve full functional recovery. Further optimization of AFG design will therefore be required to enhance its therapeutic efficacy in neural repair.

Another strategy to establish directional topological structure in hydrogels is to incorporate aligned poly(ε-caprolactone) (PCL) fibers into the hydrogel matrix. For example, Golland *et al*. integrated electrospun aligned nonwoven P11-8-enriched PCL fibers (PCL/P11-8) into a photopolymerized methacrylated collagen (collagen-GMA) hydrogel (**Figure 2D**). The resulting composite hydrogel exhibited a J-shaped stress–strain response under compression, consistent with native spinal cord tissue; concurrently, the internal fiber alignment and pore architecture effectively encapsulated neural cells and promoted axonal extension oriented along the fiber direction [50]. The hydrogel and nonwoven materials were integrated *via* a layered composite process, which has clear limitations: the guidance effect is uneven, directing axonal growth only near the fiber surface, while cells farther from the fibers remain randomly distributed, failing to achieve uniform guidance throughout the three-dimensional space. To address this bottleneck, emerging 3D printing technologies have shown distinct advantages. Zhao *et al*. filled hydrogels into 3D-printed oriented PCL fiber-bundle constructs; the uniformly aligned PCL fiber bundles effectively recruited endogenous NSCs and guided neurons to extend orderly along the fiber axis, supporting spinal cord tissue regeneration and improving motor function recovery (**Figure 2E**) [51]. It should be noted that PCL microfibers typically have diameters on the micrometer scale and therefore cannot replicate the nanoscale topography of native ECM, resulting in lower precision in guiding cellular behavior compared with AFG. Moreover, PCL is an inert hydrophobic polymer lacking intrinsic bioactivity and generally requires surface modification or incorporation of bio-adaptive components to improve cell affinity. Consequently, the development of bio-adaptive hydrogels that combine excellent topographical guidance properties remains an important direction for future research. In addition, repair of SCI should not be limited to restoration of motor function. Still, it

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must also address recovery of sensory and autonomic functions, which imposes more comprehensive design requirements on the next generation of bio-adaptive hydrogels.

The design of directional fiber structures simulates the natural structure of spinal cord white matter fiber tracts, providing a template for organized axon growth and functional restoration. Currently, mainstream fabrication techniques for aligned fiber structures can be broadly classified into eight categories: electrospinning, 3D bioprinting, magnetic alignment, freeze-casting, melt spinning, molecular self-assembly, electrohydrodynamic (EHD) printing, and microfluidic spinning. These techniques exhibit significant differences in key metrics, including fiber dimensions, alignment precision, scalability, and mechanical properties.

Electrospinning enables the low-cost, scalable preparation of nanoscale aligned fibers and is considered one of the most versatile techniques for nanofiber fabrication; however, ensuring the reproducibility of pore size and distribution remains challenging [52-56]. 3D bioprinting enables the fabrication of complex, personalized three-dimensional architectures, yet its resolution and fabrication efficiency are limited [53, 57-59]. Magnetic alignment enables non-contact, *in situ* construction of oriented scaffolds, accommodating minimally invasive delivery requirements; however, introducing magnetic particles into the fabrication system raises concerns about compositional interference [60, 61]. Freeze-casting relies solely on physical ice-crystal templating, yielding excellent biocompatibility, but the resulting scaffolds exhibit relatively small pore sizes that hinder uniform cell infiltration and dispersion [53]. Melt spinning requires no organic solvent assistance and can produce high-strength continuous microfibers, which not only enhances biocompatibility but also simplifies fabrication procedures and reduces production costs, making it more amenable to large-scale manufacturing; nevertheless, its biomimetic precision is limited, rendering it difficult to recapitulate the nanofibrous microarchitecture of the native ECM [62]. Molecular self-assembly can capitalize on spontaneous molecular assembly to construct highly hydrated, native ECM-mimicking nanofibrous networks; however, this technique is difficult to implement for macroscopic fiber orientation control, and the resulting structures possess inherently weak mechanical properties [63, 64]. EHD printing can fabricate sub-micron-scale fine fiber architectures *via* layer-by-layer stacking. Still, conventional processes are primarily applicable to two-dimensional patterning and require further modification to construct three-dimensional aligned fiber

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systems [65]. Microfluidic spinning enables precise regulation of fiber dimensional parameters, yet notable technical shortcomings persist in maintaining orderly fiber alignment, improving fabrication reproducibility, and enhancing structural tailorability [66].

The anisotropic scaffolds constructed by different fabrication processes demonstrate significantly disparate efficacies in guiding directed axonal growth (**Table 1**). In practical applications for SCI repair, the optimal fabrication technique should be selected based on core criteria including biomimetic precision, delivery modality, structural tunability, and clinical translational potential. Given that single fabrication methods all possess inherent limitations and struggle to simultaneously satisfy the multi-level, high-precision biomimetic construction demands of spinal cord tissue, future efforts should focus on developing multi-technique hybrid fabrication strategies and advancing fabrication processes specifically adapted to the structural characteristics of the spinal cord.

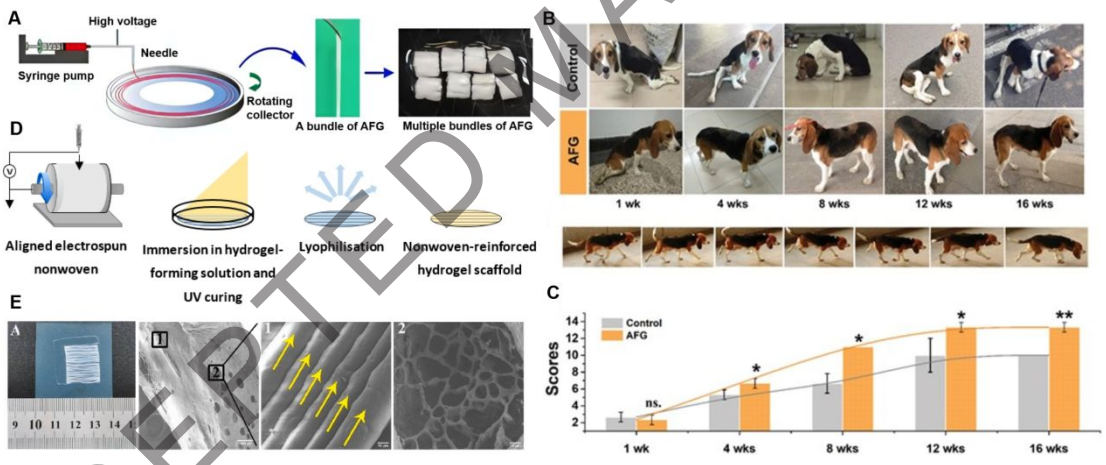


Figure 2. (A) Schematic diagram of the AFG fabrication process. Reproduced from Ref. [48] with permission of Dove Medical Press, © 2018. (B, C) Evaluation of canine spinal cord functional recovery following injury. Reproduced from Ref. [49] with permission of Oxford University Press, © 2021. (D) Fabrication of bio-adaptive scaffolds based on electrospinning technology. Reproduced from Ref. [50] with permission of MDPI, © 2022. (E) General view of the composite hydrogel scaffolds and SEM image of the composite hydrogel scaffolds (1 and 2 represent partially enlarged images with yellow arrows designating oriented fibers, and the scale is 10 μm). Reproduced from Ref.[51] with permission of Elsevier, © 2022.

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Table 1. Summary of the resolution, scalability, and effects on axon guidance of different alignment/patterning techniques.

Alignment technology	Resolution	Scalability	Effects on axonal guidance	Ref.
Electrospinning	A few nanometers to several hundred nanometers	i Minimally invasive injections can be provided. ii Oriented fiber networks can be prepared using various electrospinning devices. iii There are problems such as low mechanical strength and difficulty in ensuring the repeatability of pore size and distribution.	i Anisotropic fiber scaffolds can simulate the ECM topological structure. ii It can guide directional growth, the extension of nerve axons, and the directional migration of cells. iii The fiber diameter can be adjusted to control the guiding strength.	[52-55]
3D printing	Hundreds of micrometers	i Customizable preparation of complex three-dimensional bionic structures. ii It is necessary to explore the creation of structures that are more anatomically similar to injuries.	i It can simulate the original myelin structure and guide axons to extend along the channel. ii It has a relatively low resolution and is difficult to simulate nanoscale ECM. It is often combined with electrospinning to enhance the guiding effect.	[53, 57]

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Magnetic alignment	Nanometer level	i The external magnetic field can be remotely and dynamically regulated. ii The preparation process requires doping of magnetic particles. iii Specific conditions must be met to achieve the desired fiber orientation.	i It can guide axons to elongate along the head-to-tail axis, thereby significantly improving elongation efficiency. ii Effectively promote axon alignment and extended growth.	[60, 61]
Frozen manufacturing	Twenty micrometers to one hundred micrometers	i Directional pore structures highly similar to those of the spinal cord can be prepared. ii The aperture of the obtained scaffold is relatively small. iii Relevant research is still insufficient.	i Directional porous scaffolds can enhance axonal growth along the preferred axis. ii Microstructure regulates cell differentiation and migration. iii It significantly influences axon arrangement.	[53, 67]
Melt spinning	Micrometer level	i It can achieve large-scale production. ii It is often used in combination with electrospinning technology.	i Directional microfibers can recruit macrophages and induce their polarization towards the M2 phenotype. ii Promote Schwann cell migration, proliferation, and maturation via paracrine signaling. iii Effectively support axon extension.	[62, 68, 69]

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Self-assembly	Nanometer level	i Self-assembled fibers lack the characteristic arrangement of neural ECM. ii It needs to be used in conjunction with electrospinning technology.	i It can combine topological structure and biochemical signals to regulate the directional migration and differentiation of endogenous NSCs. ii Promotes the extension of neurites and the formation of connections between axons, effectively repairing spinal cord defects. iii Directional fibers can regulate paracrine signaling and stem cell differentiation and dynamically control the inflammatory microenvironment.	[63, 70, 71]
EHD printing	Sub-micron level	i The printing trajectory is controllable and has a promising application prospect. ii Traditional EHD technology is only applicable to two-dimensional planes. iii Improvements are required before the directional fiber structure can be constructed.	i Support long-distance nerve regeneration. ii Guide the directional growth of maturely arranged axons.	[65]

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Microfluidics	Hundreds of nanometers to hundreds of micrometers	i The fiber size can be flexibly adjusted. ii The preparation process is slow and time-consuming, making large-scale production difficult. iii Poor consistency in fiber arrangement, with limitations in repeatability and customizability.	i Oriented fibers guide neuron-like cells to be arranged along the fiber axons. ii Promote cell adhesion, proliferation, and differentiation. iii Enhance the transmission efficiency of electrical signals in three-dimensional matrices.	[66, 72, 73]

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3.2 Conductivity-enhanced designs promote electrical signal conduction in SCI repair

Electrical signal conduction is the key physical basis for the CNS to perform complex physiological functions. As the core functional carrier of nerve impulse transmission, the structural scaffold for communication and interaction between neurons and effectors within the spinal cord, and the effectors within the spinal cord are highly dependent on the integrity of the electrophysiological microenvironment [74, 75]. The physical fracture after SCI directly cuts off this electrical continuity, leading to functional impairment. To achieve efficient and stable interaction with neural tissues, an ideal electrical stimulation neural interface needs to meet multiple performance requirements. It should not only have excellent electrical conductivity and low interface impedance to ensure efficient transmission of electrical signals, but also have sufficient charge storage or injection capacity to support safe and effective electrical stimulation. Meanwhile, their mechanical properties should be well-matched to neural tissue to minimize mechanical damage and immune rejection after implantation. Bio-adaptive hydrogels with high electrical conductivity can meet the above requirements [76, 77].

Conductive bio-adaptive hydrogels exhibit mechanical properties similar to those of neural tissues. By simulating the electrophysiological microenvironment of neural tissues, they can effectively activate endogenous signaling pathways, thereby directly mediating physiological behaviors such as adhesion, proliferation, differentiation, and axonal extension in neurons and NSCs [78, 79]. The realization of its electrical properties mainly relies on the introduction of conductive materials, which can increase charge density, reduce polarization impedance, and improve voltage response. While achieving uniform and efficient electrical stimulation, it reduces the risk of cell damage from high current, provides a stable electrochemical microenvironment for cells, and ultimately enhances the metabolic and proliferative activities of nerve cells [76]. Du *et al.* introduced Zn@EGCG-modified MXene nanosheets into hydrogels. They found that as the nanosheet concentration increased, the hydrogel impedance decreased monotonically and its conductivity increased accordingly. A low-impedance hydrogel facilitates efficient electrical signal transmission and rapid electron transfer within the neuronal communication frequency band, thereby increasing the speed of bioelectrical signal conduction and enhancing information transfer to cells [80]. In another study, the conductive polymer polypyrrole (PPy) was incorporated into the hydrogel, thereby

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increasing its conductivity and reducing impedance, enabling it to exhibit conductivity comparable to that of the spinal cord and good electron-transfer behavior. This hydrogel provides a biocompatible environment that promotes endogenous nerve regeneration rather than glial fibrosis [35]. Hydrogels incorporating conductive materials can respond to externally applied electric fields, thereby promoting electrically responsive cell growth and proliferation. The amplitude of the electrical stimulation current is positively correlated with PC12 cell proliferation, and the conductive properties of the hydrogel can lower the electrical stimulation threshold, allowing cells to maintain a high survival rate at a high current amplitude of 2.50 mA. However, traditional two-dimensional conductive substrates will undergo significant apoptosis under these conditions [76]. In addition, the frequency and duration of electrical stimulation are closely related to cell responses. Low-intensity, continuous electrical stimulation that adapts to the physiological characteristics of the spinal cord can be combined with a conductive hydrogel to create a bionic electrical microenvironment, effectively promoting the differentiation of NSCs into neurons and the reconstruction of injured neural pathways (**Figure 3A**) [80]. However, there are significant differences in the electrical properties, biocompatibility, and mechanical stability of various conductive materials, which directly affect the interaction between the hydrogel and spinal cord tissue.

3.2.1 Carbon nanotube-based conductive hydrogel for SCI repair

Carbon nanotubes (CNTs) are rolled graphene sheets with a single-walled (SWCNT) or multi-walled (MWCNT) structure [81]. Their size matches that of neurons. They have high conductivity, high strength, high durability, and good flexibility. They also exhibit non-biodegradable properties and can remain stable in damaged tissues for a long time. They have strong therapeutic potential for promoting axonal regeneration and repair after SCI [82]. Studies have shown that a CNT substrate can promote neuronal and axonal growth, and this effect is related to the material's conductivity [83]. CNTs can also promote synapse formation and nerve fiber extension, enhance neural circuit connectivity, and further improve nerve cell proliferation, migration, and intracellular connectivity [84, 85]. External electrical stimulation synergizes with the biological effects of CNTs. Combining an oriented CNT/GelMA hydrogel with external electrical stimulation can further amplify the guiding effect of electrical signals, enhance the directional adhesion of PC12 cells, and promote the neuronal

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differentiation of NSCs and the directional growth of axon-like neurites [86]. In the rat T9 spinal cord transection model, this combined system effectively enhanced myelin regeneration in the injured area, ultimately restoring the rat's motor function. However, CNTs are hydrophobic and prone to aggregation and cytotoxicity at injury sites. It can cause chronic inflammation, impair mitochondrial function, increase intracellular ROS levels, and aggravate oxidative stress [87].

Physical dispersion or modification by grafting hydrophilic moieties or functional groups is a common strategy to improve CNT dispersion. In previous work, Sang *et al*. used the nonpolar segments of the amphiphilic crosslinker poly(ethylene glycol) diacrylate-dodecyl-1-(2-aminoethyl) piperazine (PEGDA–DD–AEP) to stabilize CNTs and prevent their aggregation. When combined with poly(N-isopropylacrylamide) (PNIPAM) to form a hydrogel, this material, under coupled *in vitro* electrical stimulation, promoted the growth of SH-SY5Y cells and neurites. Furthermore, in an SCI model, this hydrogel enhanced neural tissue regeneration and reduced scar formation [88]. However, because this approach relies solely on hydrophobic interactions to disperse CNTs, the physical binding is relatively weak and prone to desorption, potentially leading to reaggregation. By contrast, Vasconcelos *et al*. employed the bio-adaptive compound myricetin as a functionalizing agent and covalently anchored it to the CNT surface. This strategy not only addresses hydrophobic aggregation at its core but also integrates drug functionality with material modification. The approach obviates additional drug-loading steps, endowing the hydrogel with the electrical conductivity and structural properties of CNTs, as well as the anti-inflammatory and neuroprotective activities of myricetin [89]. When such functionalized CNTs were incorporated into a photocrosslinkable hydrogel for *in situ* SCI treatment, the hydrogel exhibited favorable safety characteristics and effectively promoted neuronal growth. Nevertheless, the functionalization process may compromise the structural integrity of CNTs, and incorporating functionalized CNTs can prolong the hydrogel photocuring time.

Therefore, developing novel CNT functionalization strategies that balance structural integrity with efficient curing is crucial for advancing their clinical translation. Against this background, polysaccharide-based molecular design holds great promise as a breakthrough approach. As a class of natural polymers rich in carboxyl, amino, and hydroxyl groups and inherently biocompatible, polysaccharides are promising

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candidates for multi-crosslinking modification with CNTs [90]. In the future, constructing highly efficient, polysaccharide-mediated CNT surface engineering strategies will effectively circumvent the drawbacks of traditional modifications, thereby significantly expanding the application boundaries of CNT-based conductive hydrogels in SCI repair.

3.2.2 Graphene-based conductive hydrogel for SCI repair

Graphene is a two-dimensional carbon nanomaterial formed by a single layer of sp^2 hybridized carbon atoms arranged in a hexagonal honeycomb lattice [91]. Graphene has super strength, light weight, flexibility, and excellent electrical conductivity, and is the preferred material for enhancing the electrical and mechanical properties of hydrogels [92]. Graphene can increase cell adhesion, promote directional differentiation of human neural stem cells (hNSCs) into neurons, and reduce glial cell generation [93]. Graphene can not only impart conductivity to hydrogels but also enhance their mechanical properties and porosity, thereby promoting uniform cell distribution and interconnection, which is necessary to accelerate axon regeneration [94]. For example, Agarwal *et al*. used amino-functionalized graphene as a crosslinking agent to prepare a graphene-collagen cryogel. The introduction of graphene significantly improved the orderliness and degree of cross-linking of collagen molecules, providing the gel with sufficient porosity and permeability within the gaps. This graphene-collagen gel promotes the migration and differentiation of BMSCs under electrical stimulation at 100 mV/mm [95]. Notably, this graphene-collagen cryogel can also control neuroinflammation and accelerate axon regeneration [96]. However, the hydrophobicity of graphene limits its dispersion in biological media. To address this, Correia *et al*. employed a 1,3-dipolar cycloaddition (DCA) approach to covalently functionalize graphene, yielding functionalized graphene (f-EG). This method improves graphene stability while largely preserving its electronic properties, making it a more suitable filler for conductive hydrogels. Conductive hydrogels prepared by combining f-EG with hyaluronic acid exhibit favorable electrical conductivity and mechanical adhesion; their mechanical properties can be matched to neural tissue regeneration, and they effectively enhance cell adhesion, viability, and differentiation, demonstrating promising potential for SCI repair [97].

Research evidence indicates that the biological effects of graphene-based nanomaterials are essentially determined by their physicochemical properties, including transverse

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size, lamellar thickness, oxidation degree, dispersion, surface charge, and dimensional structure [98-100]. However, the existing literature has yet to reach a unified consensus on this precise structure-activity relationship. The typical mechanisms by which graphene nanomaterials induce cytotoxicity include oxidative stress, mitochondrial-mediated apoptosis, cell lysis, autophagy, and necrosis [99, 101]. The key point is that the cytotoxicity of graphene is dose-dependent; therefore, establishing scientifically grounded clinical safety dosage standards is critical for its translational application in SCI treatment. However, the formulation of clinical safety dosage standards should not be limited solely to determining concentration thresholds. Instead, dosage control must be built on optimizing physicochemical properties, with the material's physical specifications first strictly confined to a safe window. Compared with nanoribbons, graphene in sheet-like configurations exhibits lower toxicity and is better suited for neural repair applications.

Furthermore, biocompatibility can be further optimized by modulating lateral dimensions and specific surface area: small-sized graphene with a particle diameter of approximately 40 nm can effectively circumvent the mechanical puncture and structural damage to cell membranes caused by larger sheet-like structures, and a moderate reduction in the specific surface area of the material can also significantly decrease cellular injury risks [102, 103]. In addition, reduced graphene oxide (rGO) exhibits substantially lower cytotoxicity than graphene oxide (GO). Its lower content of oxygen-containing functional groups on the lower surface enables spontaneous aggregation in cell culture media, effectively reducing material attachment and coverage on cell surfaces, thereby minimizing nonspecific adsorption of biomolecules such as proteins and genes and attenuating cellular damage. Surface functionalization is a key strategy to reduce graphene toxicity further and broaden its clinical safety dosage threshold. By grafting hydroxyl, amino, carboxyl, and other functional groups onto the graphene surface, the material can be endowed with a high absolute Zeta potential value, substantially enhancing its aqueous dispersion stability and biocompatibility [101, 104]. In summary, low-toxicity graphene materials adapted for clinical SCI applications must meet multiple optimization criteria, including small particle size, reduced state, high absolute Zeta potential, and moderate surface functionalization. Only on this basis can the final safety dosage threshold for clinical translation be further defined, grounded in a high-biosafety substrate.

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Carbon-based nanomaterials have good physical, chemical, and mechanical properties, as well as high electrical conductivity. They can promote axon regeneration and neural stem cell differentiation through neural electrical signals in spinal cord tissue. In addition, they have a large surface area, which is beneficial for regulating cell responses [105]. Although carbon-based materials offer significant advantages for imparting electroactivity to hydrogels and improving their mechanical properties, their cytotoxicity, hydrophobicity, and nondegradability remain unresolved, underscoring the need for safer, naturally conductive materials.

3.2.3 Conductive hydrogel based on conductive polymers for SCI repair

Conductive polymers (CPs) are a type of organic material whose main chain contains a π-conjugated structure and is chemically/electrochemically doped to generate polarons and bipolarons to achieve electronic conductivity [106]. In recent years, CPs have been widely used in biological tissue engineering due to their simple preparation, controllable electrical properties, and ease of structural modification [107]. PPy is a common conductive polymer and has been widely used to construct conductive hydrogels. PPy has high environmental stability, good biocompatibility, and adjustable conductivity. The introduction of the hydrogel confers conductivity comparable to that of natural spinal cord white matter, thereby reestablishing the conduction path for bioelectrical signals in the injured area [108, 109]. This electrically active microenvironment has been shown to significantly promote the differentiation of NSCs into neurons while inhibiting the unfavorable differentiation of astrocytes, laying the foundation for neural circuit reconstruction [110]. More importantly, PPy can enhance cell adhesion, organization, and differentiation regardless of whether electrical stimulation is performed [111]. However, PPy exhibits insufficient mechanical flexibility and poor mechanical matching with soft tissues, making it prone to stress-induced inflammatory responses.

Therefore, when preparing PPy-based conductive hydrogels, researchers typically select naturally derived molecules with high tissue affinity as matrices or functionalize PPy to improve the mechanical properties and stability of the conductive hydrogel. For example, Zhou *et al*. used tannic acid (TA) as both a dopant and a supramolecular crosslinker to form a supramolecular network with PPy chains, yielding a conductive polymer hydrogel that combined softness, high electrical conductivity, and good biocompatibility. *In vivo* experiments showed that this hydrogel could activate

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endogenous NSCs, promote their differentiation into neurons while inhibiting astrocytic differentiation, thereby significantly enhancing motor functional recovery [112]. The hydrogel is simple and efficient to prepare, requiring no complex processes; TA performs both doping and crosslinking in a single step. It exhibits favorable conductivity that matches the physiological electrical signal range of the spinal cord, extremely low charge-transfer impedance, and a formed supramolecular network that endows the hydrogel with self-healing capability. However, its functionality is relatively singular, relying primarily on electrical and mechanical adaptation, and there remains substantial room to improve the integration of multiple bio-adaptive functions. Building on this work, Fan *et al*. introduced gelatin, a highly tissue-affinitive biomaterial, and polymerized glutathione on the hydrogel to fabricate a matrix metalloproteinase (MMP)-responsive conductive hydrogel. By tuning the concentrations of gelatin and PPy, the mechanical properties and conductivity of the hydrogel could be controlled. This conductive hydrogel responds to the post-SCI microenvironment by releasing the pro-angiogenic factor basic fibroblast growth factor (bFGF), promoting axonal regeneration and angiogenesis, and improving motor functional recovery (**Figure 3B**) [113]. Such bio-adaptive hydrogels with intelligent, stimuli-responsive drug-release properties have strong potential for clinical translation. Nevertheless, gelatin is an insulating matrix; its incorporation reduces electrical conductivity relative to purely conductive polymer hydrogels, thereby limiting the efficiency of electrical signal transmission. In addition, PPy itself has poor biodegradability, and its long-term safety upon implantation remains to be validated. Future work should focus on developing bio-adaptive conductive hydrogels that simultaneously offer high electrical conductivity, controllable degradation, and multifunctional integration.

Polyaniline (PANI) is another commonly used conductive polymer. It has the advantages of easy synthesis, low cost, and reversible conversion between conductive and insulating states. However, it has poor flexibility and insufficient cell compatibility. It is prone to swelling in physiological environments, leading to decreased conductivity, and its degradation performance is not ideal. Long-term implantation may cause chronic inflammation and pain [114]. To address these shortcomings, Liu *et al*. introduced polyamino acids with good biocompatibility, degradability, and ease of functionalization to modify PANI and prepare a composite hydrogel with good

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biocompatibility and electroactivity. After the hydrogel was implanted in a rat model of SCI and electrical stimulation was applied, the experimental results showed that astrocyte proliferation was inhibited, NSCs in the injured area differentiated into more neurons, and endogenous neurogenesis was promoted (**Figure 3C**) [115]. Behavioral evaluation showed that the electroactive hydrogel combined with electrical stimulation significantly improved motor function recovery in rats. Although these studies used short-chain tetraphenylamine covalent grafting to address conventional PANI's biocompatibility and stability issues, some limitations remain. Tetraphenylamine exhibits much lower conductivity than long-chain PANI, and covalent grafting disrupts portions of the conjugated structure, which can reduce electroactivity and conductive stability.

Currently, conductive polymer-based hydrogels still suffer from suboptimal long-term biodegradability, and their implantation often induces adverse reactions, including chronic inflammation and localized pain, thereby limiting their use in SCI repair. Therefore, future research must prioritize overcoming these limitations, with particular emphasis on improving the degradation characteristics of these materials and circumventing post-implantation complications. Resolving these constraints will be crucial for advancing the translation of conductive polymer-based hydrogels from the laboratory to the clinic and realizing their widespread application in SCI repair.

3.2.4 Conductive hydrogels based on phosphorus-based materials for SCI repair

Black phosphorus (BP), a new two-dimensional semiconductor nanomaterial, exhibits good electrical conductivity, biocompatibility, and degradability, and has become an ideal filler to endow hydrogels with electrophysiological activity [116]. Under physiological conditions, it can naturally degrade into non-toxic products, such as phosphoric acid and phosphate, with high safety [74]. In addition, the BP surface is negatively charged, which can not only load growth factors or drugs *via* electrostatic interactions, thereby realizing the dual functions of conduction and local delivery, but also effectively adsorbs cations [117]. For example, the ADBF composite hydrogel developed by Wu *et al*. uses a BP matrix to efficiently load the macromolecular drug fibroblast growth factor 21(FGF21) to achieve sustained release at the SCI site, thus reducing oxidative stress of neurons; meanwhile, BP nanosheets inhibit iron death by adsorbing iron ions in the microenvironment, and further improve the immune microenvironment for nerve regeneration [118]. Although BP has the advantages

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mentioned above, it still has some limitations. BP has poor chemical stability and is easily oxidized, leading to inactivation. BP is completely degraded in aqueous solution within several days at room temperature, and its conductive network is prone to collapse, limiting its bioavailability.

To address the aforementioned issues, Liu *et al*. modified BP nanosheets with TA to enhance their stability and subsequently incorporated them into a lauroyl-modified chitosan hydrogel matrix (LAMC). The composite hydrogel exhibited not only markedly improved electrical properties and enlarged pore sizes, but also enhanced drug-delivery capability, enabling sustained release of the loaded therapeutic agent, tazarotene [119]. *In vitro* and *in vivo* experiments further demonstrated that this conductive hydrogel effectively promoted angiogenesis and neuroregeneration during SCI repair, facilitating recovery of motor function. The incorporation of BP not only increased the hydrogel's conductivity, which is favorable for neural signal transmission, but also enhanced its biocompatibility, promoting cell adhesion and proliferation. In addition, phosphate ions generated from BP degradation can supply nutritional support to neural cells, further aiding the neuroregenerative process. These findings underscore the significant potential of BP in SCI repair. However, the work primarily focused on motor-function improvement, and its effects on sensory and autonomic functional recovery remain to be explored.

Wu *et al*. proposed a novel approach to multifunctional recovery by surface-modifying BP with polydopamine (PDA); the quinone-rich structure of PDA effectively delayed BP degradation while simultaneously improving its electrical performance. The conductive hydrogel prepared on this basis exhibited capacitive coupling and generated controllable currents *in vitro* under externally applied alternating electric fields. By activating calcium-dependent signaling pathways, these induced currents promoted the differentiation of endogenous NSCs, thereby synergistically enhancing axonal regeneration and myelin reconstruction and achieving concurrent recovery of motor and bladder functions (**Figure 3D**) [120]. However, this BP-based conductive hydrogel still depended on external electrodes to deliver the electric field. Liu *et al*. leveraged electromagnetic induction by placing BP-based hydrogels in a rotating magnetic field to autonomously generate electrical signals, thereby offering a feasible solution to the aforementioned limitations. *In vitro* and *in vivo* studies confirmed that under wireless electromagnetic stimulation, BP-based conductive hydrogels exhibit enhanced anti-

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inflammatory effects and promote the differentiation of endogenous NSCs toward neuronal lineages, ultimately facilitating the recovery of behavioral and neuroelectrophysiological functions in animal models [121]. However, this study still has certain limitations. It did not investigate the dynamic distribution and degradation of the hydrogel after *in vivo* injection, and the concentration of BP must be strictly controlled during hydrogel preparation. A high concentration of BP (≥0.9 mg mL^{-1}) is cytotoxic.

In another study, Liu *et al*. employed black phosphorus nanosheets (BPNS) as active therapeutic agents and exploited the lone pairs on phosphorus atoms to scavenge ROS, thereby reducing ROS accumulation and mitigating oxidative stress damage. They also formulated a temperature-sensitive hydrogel carrier to slow BPNS degradation and prolong its therapeutic window. Upon injection, the hydrogel undergoes rapid *in situ* gelation, forming a stable matrix at the injury site that permits sustained and controlled drug release. This thermosensitive hydrogel retains BPNS at the injury site for extended periods. It sustains efficient ROS clearance, thereby reducing lipid peroxidation, inhibiting ferroptosis and apoptosis, and ultimately providing neuroprotection that promotes recovery of motor and urinary function [122]. However, the BPNS-loaded thermosensitive hydrogel still has application limitations that constrain its clinical translation. Its long-term *in vivo* biosafety has not been evaluated, the detailed mechanisms underlying axonal regeneration and synaptic reconnection remain to be elucidated, and assessment of urinary system function has been limited to voiding behavior without analyses at the neural circuit level, making it difficult to comprehensively reflect overall recovery of urinary system function.

3.2.5 Conductive hydrogel based on MXene for SCI repair

MXene is a new type of two-dimensional transition metal carbide/nitride obtained by selectively etching 'A' atoms from the MAX phase. Its structure can be represented by $M_{n+1}X_nT_x$ (n=1-3), where M represents transition-metal elements (such as Ti, Nb, V, etc.) and X represents carbon or nitrogen. Tx represents surface terminating groups (such as -OH, -O, -F, etc.), and $Ti_3C_2T_x$ is currently the most widely studied subtype [123]. Compared with traditional two-dimensional conductive materials such as graphene and CNTs, the abundant oxygen-containing functional groups on the surface of MXene endow it with excellent dispersibility in the aqueous phase, providing a

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unique materials science basis for the construction of conductive hydrogels. In addition, MXene possesses metallic electrical conductivity, a high specific surface area, easy modifiability, degradability, and good biocompatibility. It can also eliminate endogenous ROS, which can support cell growth, and shows great application potential in the field of SCI repair [124, 125]. To address the issue of insufficient cell orientation guidance ability of the developed MXene-doped hydrogels, Cai *et al*. constructed GelMA-MXene hydrogels with a micro-groove orientation structure. Relying on MXene's conductive properties, they guided the orderly proliferation and directional growth of NSCs, thereby effectively promoting the reconstruction of neural pathways after injury [126]. However, this system relies on exogenous neural stem cell loading, which poses limitations for clinical translation. For this purpose, Zhu *et al*. proposed an SCI treatment strategy that does not require exogenous cell transplantation, constructed a GelMA-MXene composite hydrogel loaded with stromal cell-derived factor-1α (SDF-1α), and recruited endogenous NSCs to migrate to the injury site *via* sustained cytokine release. At the same time, the ROS-clearing and immunomodulatory activities of MXene are used to reshape the regenerative microenvironment, and its conductive properties are leveraged to promote the differentiation of NSCs into neurons. This scheme has achieved positive results in improving the injury microenvironment and promoting neural regeneration [127]. However, its underlying mechanism of action has not been fully elucidated. The regulatory effects of MXene nanosheets on the PI3K-Akt and calcium signaling pathways still require further cell and animal experiments to verify.

Although the above-mentioned research has achieved significant breakthroughs, the long-term application of MXene-based hydrogels still faces key bottlenecks. Due to the abundance of active sites on the surface of MXene, it is prone to oxidative degradation and structural damage in physiological environments, leading to insufficient functional stability and limiting its potential for continuous application. To address this issue, Wang *et al*. adopted a metal ion intercalation modification strategy to enhance the antioxidant capacity and electrical conductivity of MXene simultaneously. They prepared a Ti_3C_2Tx-Ce composite hydrogel (TCH) by intercalating Ce^{3+} into Ti_3C_2Tx, thereby increasing its electrical conductivity by approximately 40.6%. The ROS clearance capability has been increased by approximately two times. *In vitro* and *in vivo* experiments further confirmed that this conductive hydrogel can maintain cellular

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homeostasis and alleviate mitochondrial stress by regulating the PI3K/Akt-S6 signaling pathway, and promote axon growth and synaptic remodeling synergistically, thereby clarifying the mechanism by which the MXene-based system reshapes the regenerative microenvironment through redox regulation [128].

The modification and functionalization of MXene are directions worthy of in-depth research in the future. It is necessary to develop new surface modification strategies to further enhance the functional stability of MXene, construct MXene-based conductive hydrogels with high stability and multifunctionality, and promote their clinical application in SCI repair.

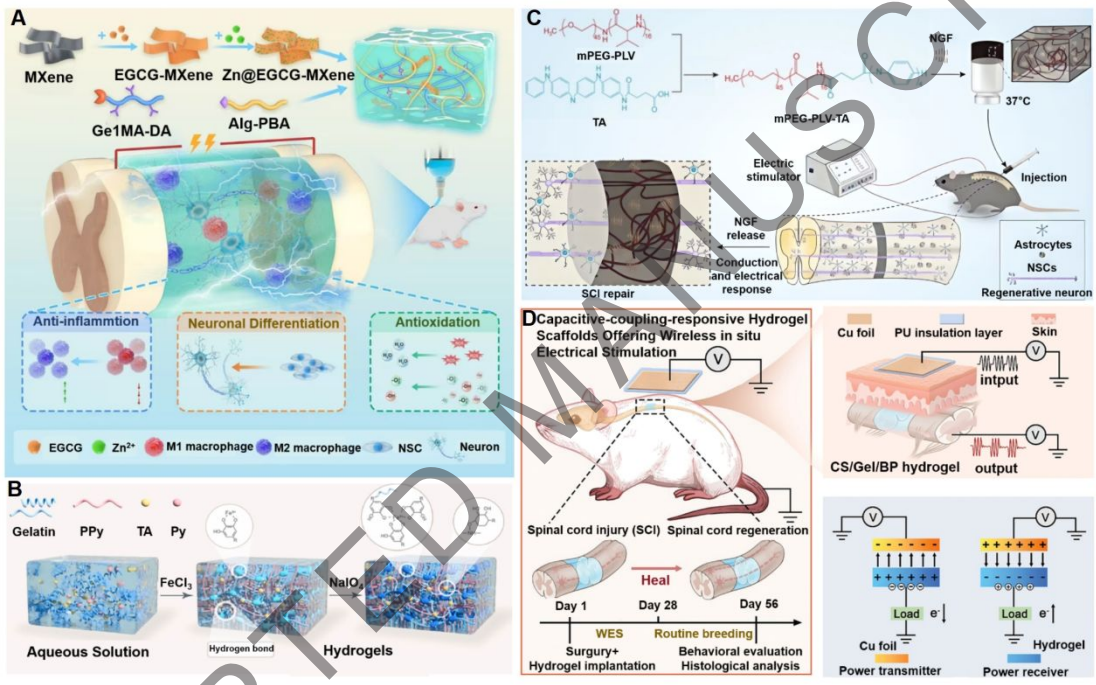


Figure 3. (A) Preparation and therapeutic role of Zn@EGCG-MXene hydrogel. Reproduced from Ref. [80] with permission of Springer Nature, © 2025. (B) Schematic diagram of the preparation of glutathione-based hydrogel. Reproduced from Ref. [113] with permission of Elsevier, © 2022. (C) Schematic illustrating electroactive hydrogel-based functional electrical stimulation for SCI therapy. Reproduced from Ref. [115] with permission of Springer Nature, © 2021. (D) Schematic diagram of wireless electrical stimulation for spinal cord repair based on conductive hydrogel. Reproduced from Ref. [120] with permission of Wiley-VCH, © 2024.

To clarify the application characteristics of different conductive material-based bio-adaptive hydrogels in SCI repair, the relevant information is summarized in **Table 2**.

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Table 2. Application and characteristics of bio-adaptive hydrogels based on different conductive materials in SCI repair.

Material class	Material composition	Conductivity (S/cm) and Test conditions	Injury/culture model	Main functional assessments	Key biological effects	Limitations/Challenges	Ref.
CNTs	CNTs in GelMA	● 0.5CNT/GelMA: $9.23\pm0.1\times10^{-5}$ ● 2CNT/GelMA : $1.57\pm0.3\times10^{-4}$ ● Electrochemical impedance spectroscopy (EIS)) ● Two-electrode method	T9 complete transection	i BBB score. ii Motor evoked potentials (MEPs). iii Thermal pain test. iv Histological and electrophysiological evaluations.	i Promote directional adhesion and cell proliferation. ii Promote the directed neuronal differentiation of NSCs. iii Regulate the inflammation—microenvironment in the injury area. iv Promote axon regeneration and myelination. v Restore tissue conductivity and	i Non-biodegradable. ii It only retains the injury area without obvious acute toxicity, but the long-term biocompatibility of long-term *in vivo* retention has not been verified. iii CNTs can be directionally dispersed in hydrogels *via*	[86]

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					improve neurological function.	electrospinning, but they tend to diffuse from fibers into PBS, and the dispersion stability during *in vivo* degradation needs to be improved.	
	SWNT in PNIPAAM	• 0.1×10^{-3} • Multifunction digital four-probe tester • 37 °C	C7 dorsal column lesion	i Cell viability assay. ii Neurite outgrowth evaluation. iii Cheerio neurological score. iv Body weight monitoring. v Histological and	i Significantly enhance SH-SY5Y cell proliferation. ii Synergistic electrical signals promote neurite growth and neural tissue regeneration. iii Inhibit scar formation.	i Need two close intercourse ice bath and ultrasound to stabilize dispersed SWNT, a cumbersome preparation step. ii Did not reveal the SWNT concentration	[86, 88]

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				immunofluorescence analyses.		or the decentralized state regulation of low electrical conductivity. iii The long-term *in vivo* degradation, the SWNT metabolic pathway, and the biocompatibility of long-term implantation were not studied.	
Graphene	Amino-functionalized graphene in collagen cryogel	• 0.1% w/v graphene-collagen cryogel: 2.91×10^{-3} • 0.5% w/v graphene-	Spinal cord organotypic slice culture	i Cell viability assay. ii Stemness gene expression analysis.	i Maintain the stemness and proliferation of (BM-MSCs). ii Promote the differentiation of	i The regulation window for graphene concentration is relatively narrow.	[95]

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		collagen cryogel: 3.93×10^{-3} ● Four-point probe method ● In PBS		iii Neuronal differentiation evaluation.iv ATP secretion detection. iv Immunomodulatory and macrophage polarization assessment. v Histological and morphological analyses.	BM-MSCs into neurons. iii Regulate macrophage polarization. iv Enhance the immunosuppressive function of BMSCs. v Support the growth and migration of spinal cord tissue cells.	ii Lack of *in vivo* validation data. iii Insufficient verification of neuronal functions.	
	f-EG in HA-Cat	● HA-Cat: $8.6 \pm 2.7 \times10^{-6}$ ● HA-C/50% EG: $1.0 \pm 0.4 \times10^{-3}$ ● HA-C/50% f-EG: $2.0 \pm 0.9 \times10^{-3}$ ● Constant voltage (0.5 V) ● Digilent Waveform	Ex vivo rat spinal cord tissue adhesion model	i Cell viability assay. ii Metabolic activity (ATP) detection. iii Neuronal marker expression. iv Histological and	i Enhance cell activity and proliferation capacity. ii Promote cell adhesion and neuronal marker expression.	i The dispersion of materials depends on the degree of functionalization. ii Lack of ability to induce neuronal differentiation.	[97]

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		Generator and UNI-T UT39 multimeter ● Calculated by Ohm's law		v Adhesion, self-healing, and injectable property evaluation.		iii Not verified for *in vivo* effectiveness. iv Long-term risks have not been evaluated.	
	PPy in Collagen (Col)	● Col/PPy40 : 0.176 ± 0.07 ● Four-probe instrument ● Room temperature	*In vitro* NSCs culture model	i Cell proliferation assay. ii Cell viability assessment. iii Immunocytochemical analysis. iv Gene expression analysis. v Protein expression analysis. vi Neurite outgrowth and	i Promote the proliferation and survival of NSCs. ii Regulate the directional differentiation of NSCs. iii Enhance neurite growth.	i The concentration control range of PPy is narrow. ii Lack of *in vivo* validation data.	[110]

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				branching evaluation.			
	PPy in TA	• TA-PPy-1：0.05 • TA-PPy-2：0.18 • TA-PPy-3：0.12 • Two-probe I–V method • Room temperature • In PBS	T9–T10 spinal cord hemisection	i BBB locomotor score. ii Footprint analysis. iii Histological staining. iv Immunofluorescence staining. v Western blot analysis. vi Neural stem cell differentiation assay *in vitro*.	i Directional regulation of NSCs differentiation. ii Inhibit inflammation and glial scar formation. iii Activate endogenous neurogenesis. iv Restore motor function.	i Long-term insufficient *in vivo* degradation and toxicity assessment. ii The ability of CPHs to maintain electrical conductivity after long-term implantation *in vivo* has not been fully verified.	[112]
PPy	PPy in gelatin	• GCH (10% PPy): 3.75×10^{-4} • GCH (20% PPy): 6.63×10^{-4} • GCH (30% PPy): 1.29×10^{-3}	T10 complete spinal cord transection	i BBB locomotor function score. ii Immunofluorescence staining.	i Inhibit inflammation and scar formation. ii Promote axon regeneration.	i The conductive mechanism is unclear, and the interface signal transduction mechanism	[113]

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		• Four-probe van der Pauw method • Room temperature • In PBS		iii Masson's trichrome staining. iv Western blot analysis. v RNA-seq analysis. vi NSC proliferation assay.	iii Enhance angiogenesis. iv Support the survival and proliferation of NSCs. v Improve the recovery of motor function.	between CPs and spinal cord tissue remains incompletely elucidated. ii Long-term *in vivo* safety has not been verified. iii The synergistic effect between exogenous electrical stimulation and material conductivity was not explored.	
PANI	PANI in methoxy polyethylene glycol-copolyvaline	• mPEG-PLV-TA(tetraniline): ≈ 10^{-4}–10^{-3} • Cyclic voltammetry and	T10 spinal cord hemisection	i BBB locomotor score. ii Footprint analysis. iii Motor evoked potentials.	i Promote axon growth. ii Promote the directional differentiation of	i The conductivity value was not directly measured.	[115]

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	(mPEG-PLV)	UV–vis verified electroactivity ● 37 °C ● In PBS		iv Immunofluorescence staining. v Histological analysis. vi Transmission electron microscopy. vii NSC differentiation and neurite outgrowth assay	NSCs into neurons. iii Inhibit the formation of glial scars. iv Activate endogenous neurogenesis. v Promote myelin regeneration and axon repair. vi Reduce the cystic cavities in the injury area. vii Restore motor function and nerve conduction.	ii The refinement of electrical stimulation parameters is insufficient.	
BP	BP in hyaluronic acid	● ADBF 0.5% hydrogel: 0.002 ● Hydrated state conductivity measurement	T9–T11 spinal cord complete transection	i BBB locomotor score. ii Footprint and hindlimb	i Efficiently adsorb iron ions and inhibit ferroptosis.	i The mechanism of action of BP has not been fully clarified. ii The BP function is	[118]

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		● Room temperature		kinematics analysis. iii Electrophysiology. iv Immunofluorescence staining. v Histological analysis. vi Mitochondrial function and ferroptosis evaluation. vii Microglia polarization (M1/M2) assessment. viii Western blot and qPCR analysis.	ii Eliminate ROS and resist oxidative stress.	masked after FGF21 is loaded. iii The long-term degradation rate and long-term biodistribution of BP *in vivo* have not been systematically investigated.	
	BP in LAMC	● LAMC/β-CD-AOI[T]/BP@TA : ≈2.0×10^{-4}	T8–T10 complete spinal cord transection	i BBB locomotor score.	i Promote angiogenesis and endothelial cell function.	i The characterization of electrical	[119]

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		• Four-probe method • Room temperature • Hydrogel samples in hydrated state		ii Footprint analysis. iii Immunofluorescence staining. iv Histological analysis. v Western blot analysis. vi Cell viability assay. vii Angiogenesis evaluation.	ii Promote nerve regeneration and recovery of motor function.	conductivity is insufficient. ii The *in vitro* effects on conduction have not been verified. iii Lack of BP concentration optimization. iv The *in vivo* experimental period is relatively short. v The problem of glial scar formation remains unsolved.	
	BP in chitosan (CS) and gelatin (Gel)	• CS/Gel/BP (BCH): 2.89×10^{-4} • Four-probe method	T9 complete spinal cord transection	i BBB locomotor score.	i Promote the proliferation of NSCs and the differentiation of directed neurons.	i The functional verification of a single BP is lacking.	[120]

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		● Room temperature		ii Footprint coordination analysis. iii Bladder weight and morphology evaluation. iv Histological staining. v Immunofluorescence staining. vi Neurotrophic factor secretion assay. vii NSC proliferation and differentiation analysis.	ii Promote the secretion of neurotrophic factors. iii Activate endogenous neurogenesis and calcium-dependent signaling pathways. iv Promote myelin repair and axon regeneration. v Reduce the cystic cavities in the injury area. vi Restore exercise and bladder function.	ii The concentration dependence of electrical conductivity was not explored. iii Only partial function recovery has been achieved. iv The problem of glial scar formation remains unsolved.	
	BP in chitosan(CS)	● BP@Hydrogel: 0.0275	T9 complete spinal cord transection	i BMS locomotor score.	i It alleviates the early	i High concentration of BP (≥0.9 mg	[121]

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	and silk fibroin(SF)	• Four-probe resistivity method • Room temperature		ii Electrophysiology. iii Thermal pain test. iv Bladder detrusor muscle morphology analysis. v Immunofluorescence staining. vi Histological analysis. vii Western blot and qPCR analysis. viii NSC proliferation and neuronal differentiation evaluation.	inflammatory response. ii Activates the PI3K/AKT pathway to regulate NSCs differentiation iii Improves motor, electrophysiological, thermal sensory, and bladder functions.	mL^{-1}) is cytotoxic. ii The visualization of dynamic distribution and degradation in hydrogels has not been achieved. iii The specific current data for the hydrogels' *in vivo* signal conduction were not detected. iv The long-term *in vivo* stability and clinical translation of radio stimulation still need to be verified.	

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MXene	$Ti_3C_2T_x$-Ce in oxidized hyaluronic acid (AHA) and dihydrazide-modified gelatin (Gel-ADH)	• $Ti_3C_2T_x$-Ce:4433 • Four-probe resistivity method • Room temperature	T9–T10 complete spinal cord transection	i BBB locomotor score ii Hematoxylin and Eosin (H&E) staining iii DHE staining iv Nissl staining v Double immunofluorescence staining vi Western blot and qRT-PCR analysis	i Antioxidant stress ii Inhibit the formation of glial scars iii Promote the regeneration of neural structures iv Improve motor function	i Long-term behavior *in vivo* has not yet been verified ii It only involves the recovery of motor functions iii The specific mechanisms underlying the pathways of different nerve cells have not yet been clarified.	[128]

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3.3 Dynamic biological responsive bio-adaptive hydrogel improves the damaged microenvironment

After SCI, a series of pathological changes, including elevated oxidative stress, microenvironment acidification, and overexpression of inflammatory factors and proteases, occur. Traditional hydrogels are difficult to dynamically adapt to changes in the damaged microenvironment. Stimulus-responsive hydrogels are a type of "intelligent hydrogel" that can sense endogenous microenvironmental signals or exogenous physical signals and respond accordingly, enabling dynamic adaptation of scaffold structure, drug-controlled release, and biological activity, thereby actively regulating the SCI repair process [129]. According to the source of stimulus signals, they can be divided into three categories: internal stimulus response type, external stimulus response type, and dual stimulus response type.

3.3.1 Internal stimulus-responsive bio-adaptive hydrogel in SCI repair

Internal stimulus-responsive hydrogels use the physiological environment of SCI lesions as the trigger to achieve targeted responses and in situ regulation without external devices. High levels of ROS, MMPs, and hyaluronidases in the local area after SCI provide natural targets for intelligent responses [130-132]. ROS-responsive hydrogels are typically constructed using dynamic chemical bonds, such as phenylborate esters. In the oxidative stress microenvironment, these dynamic bonds break, precisely releasing bio-adaptive peptides, exosomes, or anti-inflammatory drugs, eliminating free radicals and improving the damaged microenvironment. As shown in **Figure 4A**, the ROS-responsive hydrogel loaded with Apelin-13 can not only effectively eliminate ROS but also promote the polarization of macrophages to the M2 type and inhibit the M1 type phenotype, alleviate neuroinflammatory responses, and thereby improve motor function [133]. However, the functional scope of this bio-adaptive hydrogel is relatively narrow: it primarily focuses on ROS scavenging and modulation of neuroinflammation, without addressing other key reparative processes, such as glial scar mitigation, vascular regeneration, and synaptogenesis. Consequently, it inadequately addresses the multidimensional, long-term regenerative needs following SCI. Cao *et al*. fabricated an ROS-responsive hydrogel encapsulating mesenchymal stem cell (MSC)-derived extracellular vesicles (3EVs) and dexamethasone (Dxm), which, in addition to alleviating oxidative stress and inflammation, synergistically

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promoted angiogenesis and neurite outgrowth, yielding significant improvements in motor and sensory recovery **(Figure 4B)** [134]. Nonetheless, the restorative outcomes remain substantially inferior to native tissue, and achieving durable, personalized, and adaptable therapies remains challenging. Future studies should therefore pursue ROS-responsive bio-adaptive hydrogels that address these limitations.

MMP expression is upregulated after SCI, leading to destruction of the BSCB and exacerbating secondary SCI [135]. Studies have reported that inhibiting MMP-2 and MMP-9 activity protects neurons. Therefore, designing MMP-responsive hydrogels has become an important strategy for treating SCI. For example, Fan *et al*. designed a gelatin-based conductive hydrogel that responded to MMP activity and controlled the release of bFGF in the SCI microenvironment *via* glutathione coupling. On the one hand, this mechanism improves alignment between bFGF release and injury progression; on the other hand, it inhibits MMP enzymatic activity and reduces the risk of secondary injury [113]. *In vivo* experiments show that this hydrogel can promote axonal regeneration and angiogenesis and improve motor function recovery after SCI. However, it is only sensitive to elevations in MMP-2/9 and cannot simultaneously detect other critical SCI microenvironmental changes, such as ROS, pH, and inflammatory cytokines. Consequently, its capacity for adaptive responses across injury stages (acute, subacute, and chronic) remains limited, underscoring the need to develop multi-responsive hydrogels to achieve more precise, stage-adaptive repair strategies.

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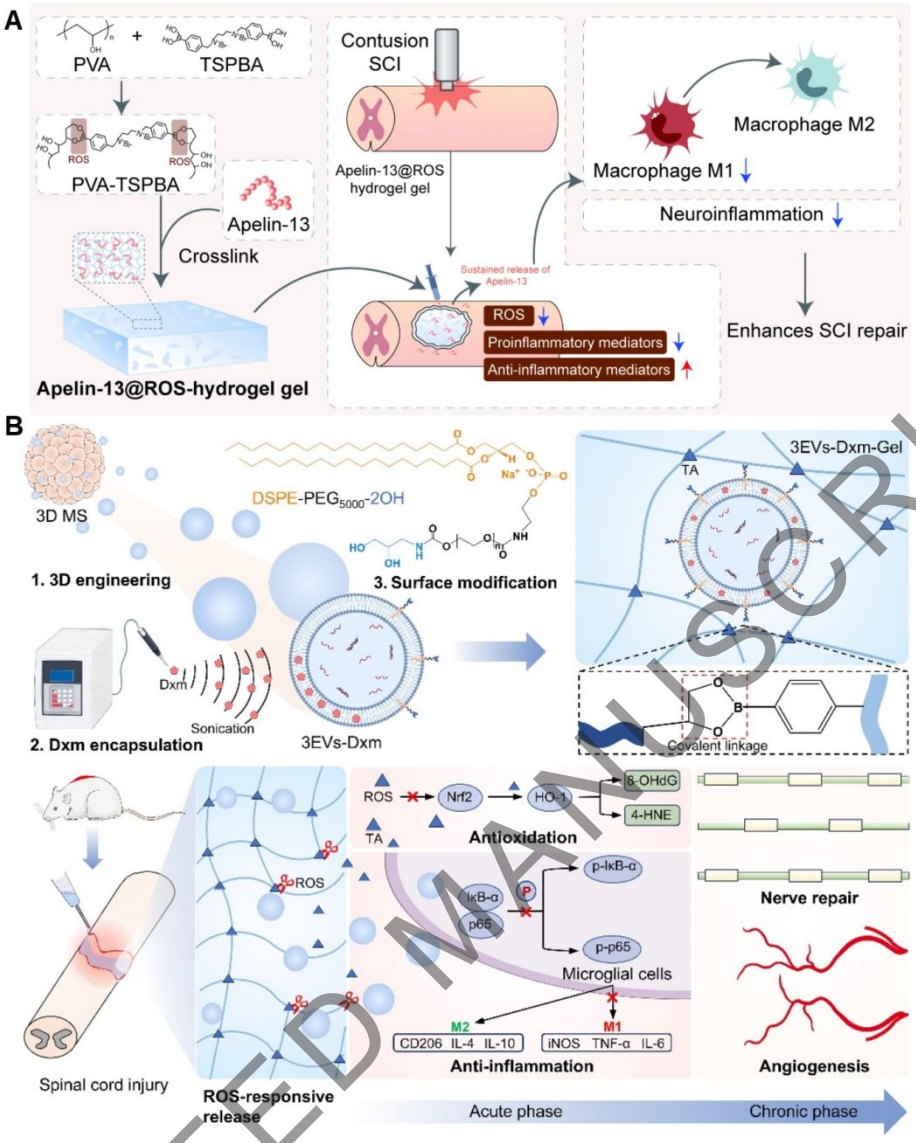


Figure 4. (A) The schematic diagram of the effect of Apelin-13@ROS-hydrogel on SCI recovery. Reproduced from Ref. [133] with permission of Springer Nature, © 2025. (B) Schematic diagram of the preparation process of 3EVs-Dxm-Gels hydrogel and its mechanism in the treatment of SCI. Reproduced from Ref. [134] with permission of American Association for the Advancement of Science, © 2025.

The above internal stimulus-responsive hydrogel can be activated in the specific pathological environment after injury, thereby improving the local microenvironment and promoting repair, effectively meeting the clinical needs of SCI repair. However, the pathological process of SCI is dynamic and sequential. Relying solely on internal biological signals to regulate drug distribution and release has certain limitations. The influence of the microenvironment on the drug delivery system's release kinetics is difficult to predict and to actively regulate after administration.

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3.3.2 External stimulus-responsive bio-adaptive hydrogel in SCI repair

In contrast to internal stimulus-responsive hydrogels, those that respond to external signals, such as light, magnetic fields, and ultrasound, offer enhanced spatiotemporal control. By utilizing external conditions, these hydrogels can more effectively adapt to the dynamic pathophysiological states encountered during SCI [136]. For instance, Dai *et al*. developed a light-responsive hydrogel with an egg-like structure, in which zeolite imidazolate framework 8 (ZIF-8) is the "egg white", paclitaxel is the "egg yolk", and TA/Fe^{3+}/tetradecanol is the "eggshell". As depicted in **Figure 5A**, the hydrogel sequentially releases drugs under near-infrared light and acidic pH conditions, enabling targeted therapy at precise time intervals [137]. This stimulus-responsive drug delivery system effectively promotes the growth of new neurons, the regeneration of mature neurons, and the formation of axons and myelin sheaths, reduces scar tissue, and accelerates the recovery of motor function. Such SCI repair should not focus solely on motor function recovery; restoring sensory and autonomic nervous system function is equally critical. Zhang *et al*. developed an ultrasound-responsive hydrogel system specifically designed to address this issue, achieving effective restoration of sensory and autonomic functions. This system utilizes a collagen–genipin biohydrogel as a scaffold and incorporates Li-PDA@ZnO nanoparticles, in which PDA-coated lithium is immobilized on the surface of ZnO nanoparticles. These nanoparticles can be remotely and wirelessly activated by ultrasound to trigger endocytosis, enabling precise spatiotemporal regulation that induces NSCs to differentiate into neurons. In addition, the hydrogel reduces local inflammation and increases the proportion of functional neurons, thereby significantly promoting the recovery of motor, sensory, and autonomic functions following SCI (**Figure 5B**) [138]. This ultrasound-responsive hydrogel offers a novel approach for treating SCI, but it remains amenable to optimization. Future work should elucidate the molecular mechanisms by which ultrasound regulates neural stem cell differentiation; establish a multidimensional functional assessment framework encompassing neurogenic bladder function, myelin regeneration, and other relevant endpoints; and refine material implantation techniques, such as conferring thermosensitivity or injectability to enhance clinical applicability.

External-stimulation-responsive hydrogels can be activated solely by external stimuli, eliminating the need for implanted devices. This spatiotemporal controllability offers a

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flexible approach to dynamic intervention in nerve regeneration. However, such strategies typically rely on complex external apparatuses that impose high operational demands and are limited in portability, thereby constraining their translational potential to some extent.

3.3.3 Dual-stimulus-responsive bio-adaptive hydrogel in SCI repair

The pathophysiological process of SCI exhibits remarkable temporal dynamics, sequentially progressing through acute, subacute, and chronic phases, with fundamental differences in microenvironmental characteristics, injury mechanisms, and repair requirements across these stages. The acute phase is characterized by secondary injury cascades as the core pathological feature, in which massive accumulation of ROS and excessive inflammatory responses continuously damage residual neurons [139]. The primary therapeutic goal during this stage is ROS scavenging, inflammation suppression, and interruption of secondary injury cascades, ROS-responsive hydrogels can effectively address. The subacute phase represents a critical transitional period from injury to tissue repair, with therapeutic priorities focusing on remodeling the pro-repair immune microenvironment, suppressing excessive glial scar proliferation, and promoting axonal and vascular regeneration [3, 8]. Bio-adaptive hydrogels with oriented topological structures and MMP-responsive hydrogels can fulfill these objectives. Upon entering the chronic phase, the pathological status of SCI stabilizes; however, the formation of lesion cavities, consolidation of glial scar barriers, and difficulties in neural circuit reconstruction become major bottlenecks for repair [140, 141]. Conductive hydrogels can meet the demands for neural functional reconstruction during the chronic phase by remodeling the glial scar microenvironment, bridging tissue defects, and inducing long-distance axonal regeneration. The temporally dynamic pathological characteristics of SCI dictate that stage-specific hydrogels represent an imperative direction for precision therapy.

Nevertheless, the aforementioned stage-specific single-function hydrogels can only address the repair demands of a single disease stage, failing to accommodate the dynamically evolving pathological features throughout the entire disease course, thereby making precise, whole-course intervention difficult to achieve. Consequently, bio-adaptive hydrogels integrating multi-responsive properties have emerged as a research focus, with material design progressively evolving from single-function to

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temporally sequential multifunctionality. Chen *et al*. developed a ROS/MMP dual-response injectable hydrogel that uses poly(vinyl alcohol) /TSPBA (PVA/TSPBA) as the matrix and is loaded with CNT@MnO_2 nanomedicine and vascular endothelial growth factor (VEGF). In the acute phase, the hydrogel can release nanomedicines in response to ROS to scavenge them and reduce inflammation; in the later phase, it can release VEGF in response to MMPs, thereby promoting neural stem cell differentiation and vascular regeneration, ultimately facilitating effective recovery of motor function (**Figure 5C**) [142]. This dual-responsive hydrogel effectively addresses the limitations of single-responsive hydrogels: it overcomes the insufficient responsiveness and limited sustained neuroprotective efficacy of ROS-responsive hydrogels in later stages of injury, and it also transcends the constrained adaptive capacity of MMP-responsive hydrogels to acute, subacute, and chronic injury microenvironments. However, current research in this area still exhibits notable limitations. On one hand, the *in vivo* MMP-responsive stability of such hydrogels, as well as their actual therapeutic efficacy, long-term biosafety, and effectiveness following implantation, remain to be systematically validated. On the other hand, SCI is intrinsically a cascading network mediated by multiple cell types, molecules, and signaling pathways, characterized by high spatiotemporal heterogeneity. Even dual-responsive materials still represent an intervention model targeting limited pathological nodes, capable of modulating only partial pathological processes rather than systematically overcoming the overall repair bottlenecks within the SCI micro.

Abnormalities in the SCI microenvironment encompass not only biochemical regulation but also multiple dimensions, including mechanical properties, electrophysiological activity, energy metabolism, and ECM remodeling. The native spinal cord is an electroactive tissue, and nerve signal conduction, neuronal differentiation, axonal extension, and synaptic remodeling are all highly dependent on a stable electrophysiological microenvironment [143]. Simple conductive hydrogels merely introduce electrical conductivity and are unable to recapitulate the coupled multi-physical and biochemical microenvironmental characteristics of the native spinal cord. In contrast, dual-stimuli-responsive hydrogels that integrate endogenous microenvironmental signals with exogenous physical stimuli can achieve synergistic, multidimensional microenvironmental remodeling through the cooperative action of electrical signals and biochemical factors, better simulating the physiological properties

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of the spinal cord and providing favorable conditions for nerve regeneration. For instance, Fan *et al*. designed a hyaluronidase/magnetic-field dual-responsive hydrogel using hyaluronic acid-gelatin as a carrier and loaded it with Nerve Growth Factor (NGF) and Fe_3O_4 nanoparticles. In the injured area, endogenous hyaluronidase degrades hyaluronic acid, enabling targeted and sustained NGF release to promote nerve regeneration and thereby improve motor function; at the same time, Fe_3O_4 nanoparticles can respond to external magnetic fields and, when combined with magnetic stimulation, further improve the injury microenvironment [144]. Compared with the functional limitations of a single-response system, the dual-stimulation-responsive hydrogel can achieve intelligent, spatiotemporal-controlled drug release, providing more precise, controllable, and continuous drug delivery to the injured site, which is closer to the clinical demand for precise and personalized regenerative medicine. Although dual-stimulation-responsive hydrogels have demonstrated significant advantages, they can only improve motor function recovery and have no clear effect on sensory or autonomic nerve function, suggesting that their therapeutic scope remains insufficient.

In addition, the complex compositional structure of dual-stimulus-responsive hydrogels often leads to a series of problems, such as increased synthesis difficulty, reduced biological activity of the active ingredients they carry, and decreased release efficiency, which seriously hinder their clinical application. In the future, it may be possible to start with the following aspects to promote its clinical-translational nature. (i) In terms of simplifying the synthesis process, a one-step molding process based on the orthogonal gelation principle can be adopted. By using two non-interfering cross-linking reactions, dual-responsive hydrogels with core-shell or layered structures can be prepared, avoiding the complexity of multi-step embedding processes [145]. A one-step physical cross-linking process can also be adopted. For specific drugs with their own stimulus-responsive ligands (such as positively charged antibodies/proteins), they can be linked with polysaccharides or conductive polymers with opposite charges (such as PEDOT: PSS) through electrostatic complexation and hydrogen bond physical cross-linking to form injectable dual-responsive (such as REDOX/pH) hydrogels without the need for chemical cross-linking agents. It significantly reduces synthesis complexity, and the protein can maintain its natural conformation and biological function in a mild physical mixture [146]. (ii) In terms of the protection of active ingredients, a mild cross-linking system without free radicals at room temperature, such as a Schiff base reaction and

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coordination bond, is selected to replace the traditional intense conditions like ultraviolet light polymerization, thereby reducing the conformational damage and activity loss of active ingredients from the source [147, 148]. (iii) In terms of enhancing release efficiency, segregated encapsulation can be adopted. Hydrophobic or easily inactivated drugs can first be loaded onto nanocarriers such as liposomes, micelles, and PNPs, and then physically encapsulated within the influent gel network. Nanocarriers isolate drugs from gel crosslinking agents or residual monomers, reducing explosive release and non-specific adsorption. After gel swelling or degradation is triggered by stimulation, the nanocarrier sequentially releases the drug, thereby achieving stepwise or synergistic dual-stimulation-controlled release, thereby solving the problems of release blockage and low encapsulation rate[149].

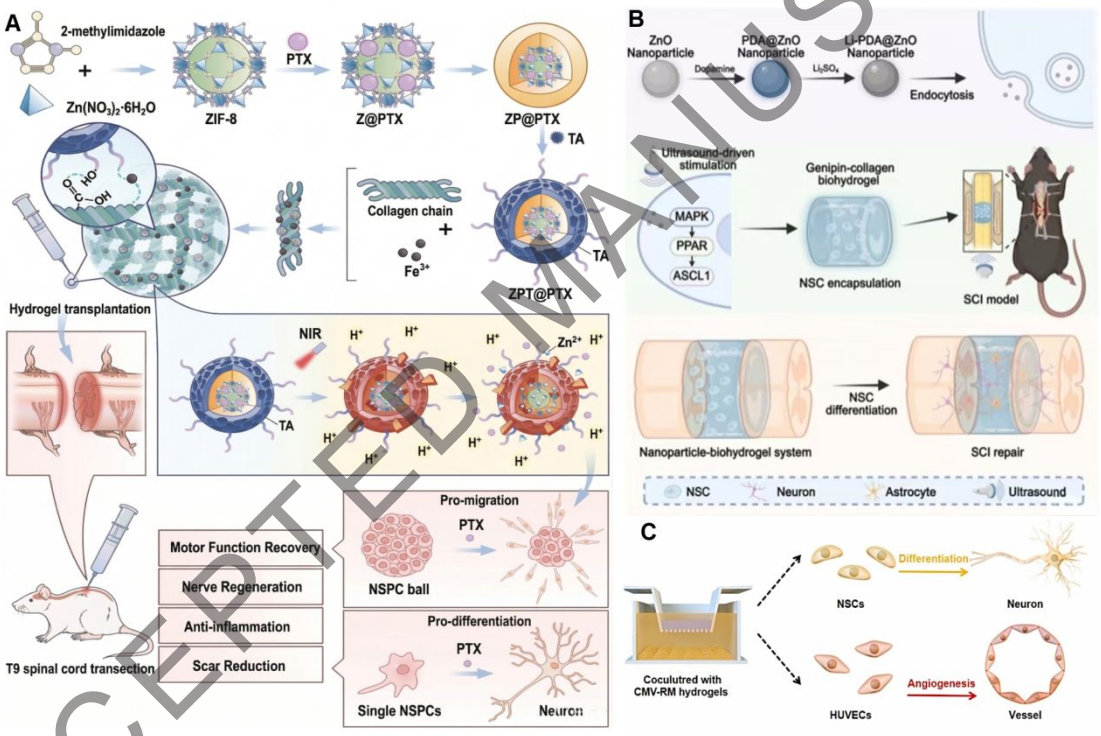


Figure 5. (A) Schematic diagram illustrating the synthesis of nanoparticles and hydrogels and the subsequent drug release. Reproduced from Ref. [137] with permission of Wiley-VCH, © 2023. (B) Schematic diagram of nanoparticle-biohydrogel enhancing the differentiation of NSCs. Reproduced from Ref. [138] with permission of Elsevier, © 2025. (C) Schematic diagram of the effect of CMV-RM hydrogel on neuronal differentiation and angiogenesis. Reproduced from Ref. [142] with permission of Elsevier, © 2025.

3.4 Interface tissue adhesive bio-adaptive hydrogel promotes neuronal connection

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Achieving tight integration with damaged tissue and accelerating nerve regeneration within complex microenvironments remain major challenges in SCI therapy. Conventional hydrogels often fail to integrate effectively and stably with native tissue, resulting in insufficient infiltration of endogenous NSCs into the transplanted hydrogel and suboptimal neural regeneration outcomes [150, 151]. Therefore, developing hydrogels with strong adhesiveness and bioactivity is of great significance for SCI repair. Enhanced hydrogel adhesiveness enables the material to form a strong bond with the lesion site, guiding newly formed neurons to bridge the severed spinal cord stumps, promoting neuronal tissue repair and achieving stable functional recovery after SCI [152]. Dopamine (DA), inspired by mussel adhesion mechanisms, is commonly used to modify biomaterial surfaces to enhance adhesion and cell affinity; it also promotes neuronal growth and differentiation, demonstrating promising potential for SCI repair [153-156]. Through covalent and noncovalent interactions, DA can mediate stable adhesion between hydrogels and tissue interfaces; however, the catechol groups in DA can inhibit free-radical polymerization and are prone to spontaneous oxidative polymerization into PDA under alkaline conditions, which can compromise the long-term adhesive stability of hydrogels (**Figure 6A**) [157]. Chai *et al*. introduced the bio-adaptive peptide isoleucine-lysine-valine-alanine-valine (IKVAV) to enhance hydrogel adhesiveness and bioactivity, offering a new strategy to improve integration of bio-adaptive hydrogels with tissue interfaces [100]. This hydrogel can effectively promote neural stem cell differentiation and angiogenesis, suppress glial scar formation, and improve neurological functional recovery after injury. However, the non-degradability of PNIPAM limits its further application. To address this, Liu *et al*. selected biodegradable SF as the hydrogel matrix, methacrylated it, and combined it with laminin-acrylate (LM-AC) and a photoinitiator (LAP). *Via* local injection and UV irradiation, an SF-GMA/LM-AC composite hydrogel can be rapidly formed at the injury site. This hydrogel firmly adheres to the spinal cord surface and effectively promotes the regrowth of myelinated axons into the lesion site after injury, thereby facilitating motor function recovery. Compared with physically adsorbed conventional hydrogels, the SF-GMA/LM-AC hydrogel exhibits superior infiltration and adhesion to spinal cord tissue; LM-AC effectively promotes neural stem cell differentiation and growth, enhancing the material's bioactivity [158].

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Nevertheless, under the action of this bio-adaptive hydrogel, some axons grow in a disorganized manner, reducing repair efficiency. To address this issue, Chen *et al*. combined electrospinning with *in situ* crosslinking to fabricate an adhesive, aligned collagen–fibrinogen (Col-FB) hydrogel; this aligned fibrous hydrogel more closely mimics the anisotropic microstructure of the native spinal cord (**Figure 6B**) [159]. Transplantation of the hydrogel into the transection site in rats achieved firm integration with the spinal cord stumps. Crucially, the hydrogel's oriented topography recruits endogenous NSPCs and guides their directed differentiation into neurons, significantly improving motor function recovery.

The above strategies to enhance hydrogel adhesiveness have mainly focused on promoting structural repair and motor functional recovery after SCI by integrating bio-adaptive hydrogels with host tissue interfaces, with comparatively less attention paid to restoring autonomic functions, such as bladder control. Researchers have attempted to address this shortfall by incorporating adhesive peptides into bio-adaptive hydrogels; however, these peptides primarily act at the cellular interface and provide limited adhesion at the tissue–implant interface, often necessitating co-use with DA. To address this, Xiao *et al*. grafted 3,4-dihydroxy-L-phenylalanine (DOPA) onto chitosan and designed the peptide HGF-RADA16-IKVAV (HRI) to prepare a double-network (CD/HRI) hydrogel. This hydrogel can achieve seamless integration with host tissue (**Figure 6C, D**) [160]. The close contact between neurofilaments and myelin sheaths, as well as remyelination, was confirmed by electron microscopy images and Z-stack images (**Figure 6E, F**). Previous studies indicate that the RADA16 component of HRI can self-assemble into a nanofibrous scaffold, and the introduction of adhesive peptides markedly enhances the hydrogel's bioactivity, creating a favorable microenvironment for neural stem cell proliferation and differentiation [161]. Moreover, the dynamically reversible π–π interactions within the CD/HRI hydrogel confer self-healing capability and increase overall mechanical strength, thereby facilitating the establishment of a stable structural bridge across the lesion and guiding orderly axonal regeneration. Crucially, this hydrogel demonstrates multifaceted reparative functions: it can modulate local immune responses, promote axonal regrowth, support the formation of neurotransmitter-associated synapses, and facilitate remyelination, thereby improving motor and sensory functions and enhancing recovery of bladder function. Although it facilitates motor recovery, its efficacy remains room for improvement, and the

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efficiency of integrating regenerated axons into host functional networks requires further optimization.

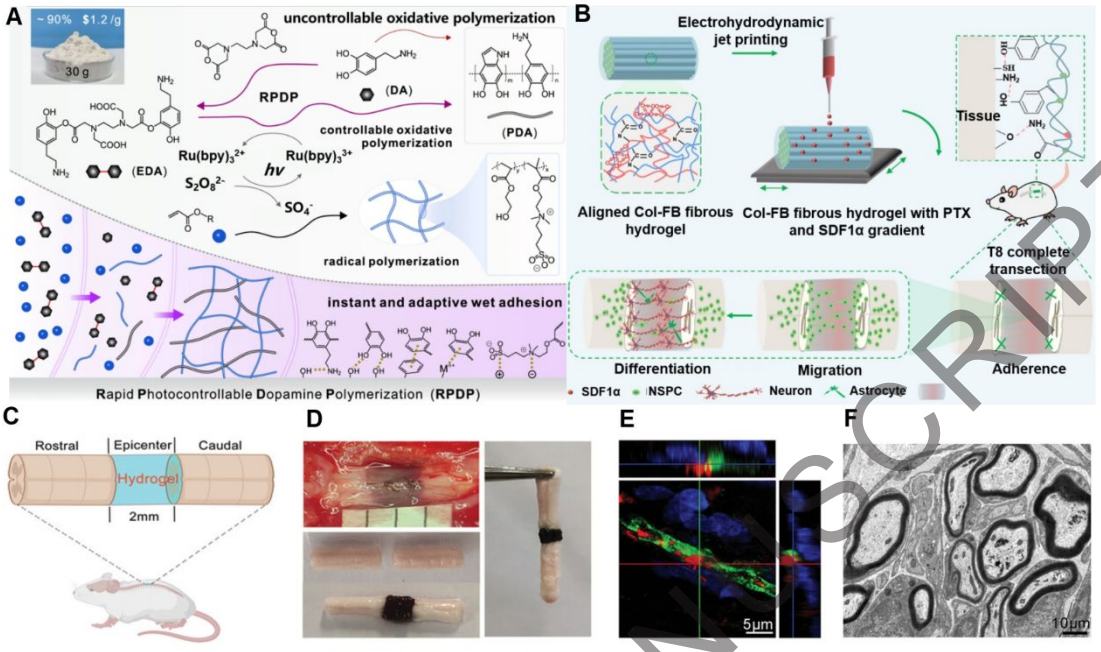


Figure 6. (A) Schematic of the *in situ* forming adhesive and bio-adaptive glycidyl methacrylated SF-GMA/LM-AC hydrogel to improve neural regeneration in SCI rats. Reproduced from Ref. [157] with permission of Springer Nature, © 2026. (B) Schematic of adhesive, stretchable, and spatiotemporal delivery-aligned Col-FB fibrous hydrogel to improve nerve reconnection in SCI rats. Reproduced from Ref. [159] with permission of American Chemical Society, © 2021. (C) Schematic diagram of the hydrogel being injected into a rat SCI site. (D) Digital photograph showing strong adhesion between the hydrogel and the spinal cord. (E) Z-stack image. (F) Ultrastructural image of the hydrogel in the cross-section of the lesion. Reproduced from Ref. [160] with permission of Wiley-VCH, © 2023.

4. Bio-adaptive hydrogels in inhibiting scar formation and improving the microenvironment after SCI

The formation of scar tissue is considered to be the main obstacle to functional recovery after SCI. After SCI, reactive astrocytes proliferate rapidly and migrate to the injury center, together with infiltrating fibroblasts, to form dense scar tissue. This kind of scar tissue has a protective effect in the early stage and is a physical barrier to limit the spread of inflammatory cascade reactions. Through physical action, it can seal the injury site, isolate the injured area from surrounding tissue, effectively reduce secondary neuronal death and oxidative stress, and maintain the integrity of the BSCB. However, in the chronic stage, chemical inhibitory factors such as CSPGs and collagen

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secreted by scar tissue activate downstream signal pathways, which not only constitute the physical boundary that hinders axonal extension, but also form a lasting chemical inhibitory microenvironment, which inhibits axonal regeneration and neural plasticity, and hinders the functional recovery of SCI [11, 162-165]. As long as the scar persists, this inhibitory chemical signal will remain, even with neural stem cell transplantation or exogenous delivery of a neurotrophic factor. It is difficult to achieve effective neural repair. Inhibition of scar formation or degradation of its components is the basis for reshaping the neural loop and ultimately promoting recovery of motor function.

4.1 Bio-adaptive hydrogel in immune regulation mediates scar formation

The imbalance of the immune microenvironment is the initial stage of excessive scar formation after SCI. By regulating the infiltration and polarization of immune cells, inhibiting the activation of pro-inflammatory M1 microglia, and promoting their transformation into the anti-inflammatory M2 repair subtype, the degree of scarring can be reduced [34, 166, 167]. Using an active hydrogel to adjust the microenvironment at the injured site can inhibit scar formation at its source. For example, DA-modified adhesive hydrogels exhibit immunomodulatory properties and modulate scar development. Previous studies have shown that removing excess ROS in the injured area can not only reduce abnormal ROS-mediated astrocyte activation but also specifically regulate the polarization balance of microglia, thereby reducing astrocyte stimulation by inflammatory factors and the densification of scar tissue (**Figure 7A**) [168]. Therefore, using ROS-responsive hydrogels to remove ROS can be an effective strategy for regulating immune responses and intervening in scar formation. **Figure 7B** illustrates the mechanism through which various types of bio-adaptive hydrogels inhibit scar formation by modulating the immune microenvironment.

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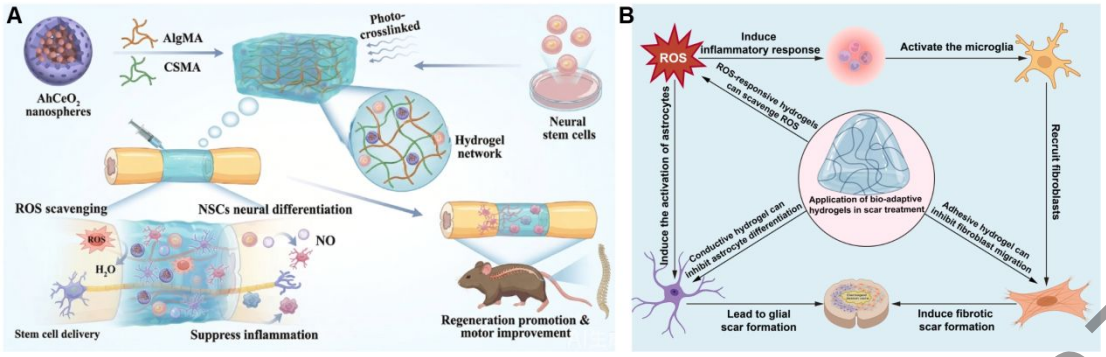


Figure 7. (A) Schematic diagram of $AhCeO_2$ nanospheres alleviating oxidative stress and remodeling the neuroinflammatory microenvironment. Reproduced from Ref. [168] with permission of American Chemical Society, © 2025. (B) The mechanism by which different types of bio-adaptive hydrogels inhibit scar formation by regulating the immune microenvironment. The figure was created using BioRender.com.

4.2 Bio-adaptive hydrogel inhibits glial scar formation

Glial scars are the main physical barrier for nerve regeneration and functional recovery after SCI [169, 170]. Its core components include reactive astrocytes and abundant deposits of scar-associated ECM. Currently, the use of bio-adaptive hydrogels to target the key pathways or core components of scar formation, by inhibiting the excessive activation of astrocytes and regulating the balance of ECM synthesis and degradation, disintegrating this physical barrier is a common strategy to inhibit glial scar formation and promote nerve repair. Different types of hydrogels can significantly inhibit glial scar formation through distinct mechanisms. For example, the β-GP/CS@ATRA composite hydrogel developed by Zhang *et al*. specifically inhibits the Pin1 protein and its downstream PI3K/AKT/CDK2 pathway by loading all-trans retinoic acid, thereby reducing reactive astrocyte proliferation and ECM secretion, such as collagen [171]. The functional self-assembling peptide (F-SAP) bio-adaptive hydrogel designed by Sun *et al*. targets the CSPG signaling pathway, uses specific peptide segments to antagonize PTPσ/LAR receptors, blocks signals that promote scar formation, and induces microglia to polarize towards the M2 phenotype, reducing inflammation-mediated ECM densification **(Figure 8A)** [172]. The bFGF/PTX@BSA hydrogel constructed by Zhu *et al*. synergistically inhibits scarring and protects nerves. Paclitaxel downregulates scar-related molecules, such as CSPGs.

In contrast, bFGF promotes axon regeneration, thereby breaking the vicious cycle in which scars inhibit regeneration (**Figure 8B**) [173]. In addition, the conductive

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hydrogel can induce directional differentiation of endogenous NSCs into neurons *via* the electrical microenvironment, inhibit their differentiation into astrocytes, and effectively prevent excessive glial scar formation [79, 174-176]. However, astrocytes themselves are an important component of the BSCB and have neuroprotective effects [164]. Excessive inhibition of glial scar formation may weaken tissue integrity, leading to loss of protective barriers and increasing the risk of cerebrospinal fluid leakage and infection. Therefore, the core difficulty in glial scar intervention lies in balancing the weakening of the regenerative inhibitory barrier with the maintenance of spinal cord tissue integrity.

A persistent inflammatory microenvironment fundamentally drives the excessive proliferation of glial scars. Compared with directly degrading or eliminating mature scar tissue, reducing pathological scar formation at its source through upstream regulation of microglial polarization and correction of metabolic disturbances in the lesion can maximally preserve the original tissue architecture of the spinal cord. Existing studies have confirmed that FBXL12 is a critical target mediating scar-free repair after SCI: upregulating FBXL12 expression *via* N6-Methyladenosine (m6A) modification can reprogram microglia toward a repair phenotype, downregulate the release of pro-inflammatory factors such as TNF-α and IL-1β, and indirectly block excessive astrocyte activation and massive CSPG deposition [177]. Animal experimental results demonstrated that following treatment, scar volume was reduced by approximately 70%, yet a continuous astrocytic boundary was retained, the integrity of the BSCB was not compromised, and inflammation did not spread to surrounding tissues; concurrently, axons could successfully traverse the scar region, and hindlimb motor function in rats was significantly restored. This indirect regulatory approach avoids the risks to tissue integrity associated with direct scar destruction. However, achieving specific FBXL12 overexpression in microglia within the injury region remains challenging. Existing AAV delivery schemes driven by the IBA1 promoter are insufficient to achieve complete scar-free repair, as the SCI microenvironment is co-regulated by multiple cell types. If this gene delivery system is combined with an adaptive hydrogel, leveraging the hydrogel's spatiotemporally controlled release and biomimetic microenvironmental regulation to act synergistically, it is expected to better balance scar inhibition with tissue-protective effects and significantly enhance the clinical translational potential of this target therapy.

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4.3 Bio-adaptive hydrogel relieves fibrotic scars

After SCI, infiltrating fibroblasts secrete a small amount of type I and III collagen, thereby forming fibrotic scars. Although axonal sprouting can extend around glial scars, its growth is often inhibited when it comes into contact with fibrotic scars [178, 179]. The essence of fibrosis is the excessive deposition of ECM, and a variety of bio-adaptive hydrogels have shown promising potential for eliminating such scars. For example, an adhesive hydrogel containing the adhesion peptide IKVAV can inhibit fibroblast migration to the injury core, block the TGF-β1 signaling pathway, reduce endoplasmic reticulum stress, and reduce the secretion of key components of fibrotic scars [180]. Another I-5/ARSB composite hydrogel specifically degrades 4-sulfated chondroitin proteoglycans (C4S) and reduces the expression of inhibitory CSPGs by loading chondroitin 4-sulfatase (ARSB). At the same time, the I-5 component inhibits TGF-β1-induced fibroblast activation and fibrosis reactions, providing a favorable physiological microenvironment for axonal growth (**Figure 8C**) [181]. However, fibrotic scars also have a certain positive effect in sealing the injured area. If scar tissue is completely removed, cavities incompatible with nerve regeneration may be left, disrupting the normal ECM network and biomechanical homeostasis [182]. Therefore, remodeling rather than simply eliminating the scar microenvironment is key to achieving long-term repair. Studies have shown that platelet-derived growth factor receptor beta-positive (PDGFRβ$^+$) pericytes migrate from the vessel wall to the injury center after injury and are the main source of fibrotic scars [183]. Recent studies have used hydrogels to directionally regulate the distribution of PDGFRβ$^+$ cells, converting fibrotic scars into a matrix that supports axonal growth and providing innovative treatment strategies to guide axonal regeneration [184].

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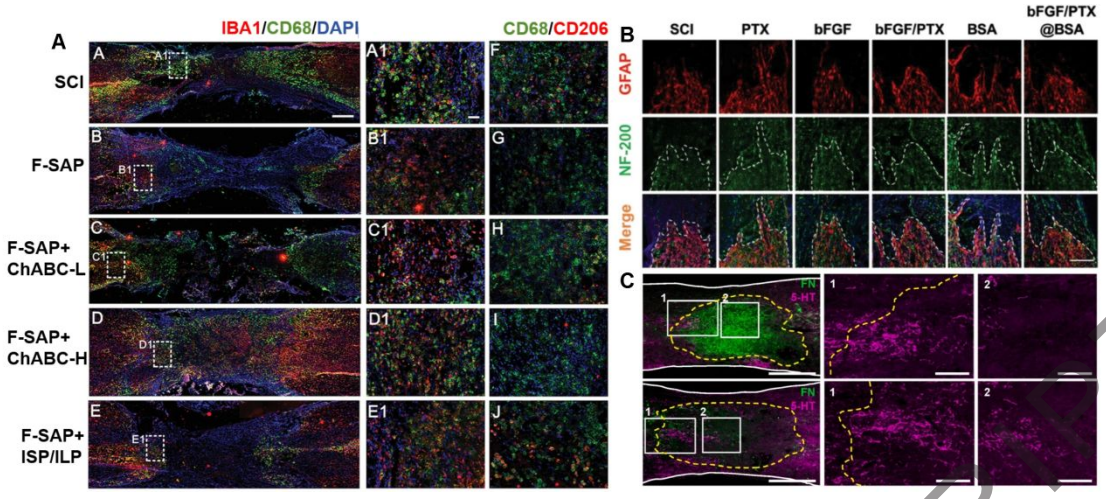


Figure 8. (A) Immunofluorescence staining images of the main markers of the anti-inflammatory microglial/macrophage activation phenotype. Reproduced from Ref. [172] with permission of Wiley-VCH, © 2023. (B) Images of the injury epicenter of the spinal cord. Reproduced from Ref. [173] with permission of Wiley-VCH, © 2024. (C) Representative images of longitudinal spinal cord sections at 5 weeks post-injury. Reproduced from Ref. [181] with permission of Elsevier, © 2022.

5. Application of bio-adaptive hydrogel in promoting motor function recovery

The spinal cord is the core hub for transmitting movement instructions between the brain and the body. SCI disrupts the structural integrity and functional connections between the higher centers and the spinal cord, damaging neural circuits and ultimately interrupting movement signals transmitted from the brain to skeletal muscles. This pathological process not only causes motor dysfunction but also causes permanent paralysis, causing the patient to lose the ability to take care of himself and become dependent on rehabilitation treatment for a long time [1, 185]. Anatomically, SCI can damage descending nerve pathways, including the corticospinal tract (CST), rubrospinal tract, and reticulospinal tract. As the core pathway that mediates voluntary movement of the body, disruption of the CST directly blocks the synaptic connection between the motor cortex and the motor neurons in the anterior horn of the spinal cord, preventing central motor signals from being transmitted downward, thus leading to the loss of limb motor function below the injured level [186, 187].

5.1 Guide the orderly extension of axons

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Cavities and glial scars are easy to form in the injured area after SCI, which hinders axon regeneration and signal conduction. The porous structure of hydrogels can serve as a structural scaffold for axons. In contrast, bio-adaptive hydrogels with directional topological guidance can provide paths for the orderly growth of axons, thereby promoting the recovery of motor function. Yao *et al*. prepared a conductive CNT/GelMA hydrogel with an aligned structure, which combined conductivity and directional guidance. Among them, the aligned fiber structure mimics the orderly arrangement of nerve axons, providing topological cues to guide directional axon regeneration and overcome the glial scar barrier. At the same time, the introduction of CNTs gives the hydrogel good conductivity, which can simulate the electrophysiological microenvironment of nervous tissue and promote nerve signal conduction. The combination of the two promotes axonal regeneration and improves motor function [86]. In another study, the double-crosslinked biomimetic composite hydrogel with a directional topological guidance structure constructed by Zhao *et al*. can not only promote the differentiation of NSCs into neurons, but also guide axons to extend in an orderly manner along the axial direction of the spinal cord (**Figure 9A**); in a rat spinal cord semi-transection injury model, the hydrogel can significantly reduce the injury cavity and significantly improve the Basso, Beattie, and Bresnahan (BBB) locomotor rating scale. The inclined plate test further showed that the left lower limb of the rat implanted with the hydrogel can stably support the body during crawling movements without obvious dragging, and grip strength is good. These results demonstrate that the bionic composite hydrogel effectively promotes tissue regeneration and motor function recovery [51]. The BBB motor function score and the inclined plate test are common methods for evaluating hind-limb motor function, weight-bearing grasping, and muscle endurance after SCI. However, the assessment scope of both is mainly limited to macroscopic movement, and their ability to distinguish fine motor and sensory functions is insufficient. Moreover, the results are determined by manual observation, which is highly subjective and difficult to reflect the true neural repair effect. Therefore, establishing a more objective and comprehensive functional recovery evaluation system is an important issue that needs to be addressed in this field in the future.

5.2 Induce endogenous nerve regeneration

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Secondary injuries after SCI (such as the inflammatory response, iron overload, and scar formation) exacerbate nerve tissue damage and inhibit endogenous nerve regeneration. Bio-adaptive hydrogels can precisely regulate the microenvironment of the injured area, remove harmful factors, create an anti-inflammatory repair environment, and reshape the conditions for nerve regeneration, thereby activating the endogenous repair mechanism and promoting the recovery of motor function.

Following SCI, the release of damage-related molecular patterns (DAMPs) can initiate an immune response, leading to the infiltration and activation of immune cells. This process creates an abnormal inflammatory microenvironment that exacerbates secondary injury and impedes recovery of neurological function [188, 189]. To address this problem, Shen *et al*. developed a dual-effect immunomodulatory hydrogel (Scav/Delv-GL) that combines the functions of DAMPs, scavenging and sustained interleukin-10 (IL-10) release. The hydrogel is based on photo-cross-linked gelatin and is modified with cationic polymer polyamidoamine (PAMAM-G3) to impart cationic properties. It is also loaded with IL-10, which can specifically bind and remove anionic DAMPs at the damaged site *via* electrostatic interactions, blocking the activation of abnormal immune responses at the source. The sustained release of IL-10 inhibits the secretion of pro-inflammatory factors, promotes macrophage polarization toward an M2 anti-inflammatory phenotype, and establishes an anti-inflammatory repair microenvironment. This hydrogel can effectively inhibit the formation of glial and fibrous scars, reduce the regenerative barrier, provide favorable conditions for endogenous NSC differentiation and axon growth, and ultimately achieve significant improvement in long-term motor function after SCI [190].

Apart from inflammatory reactions, ferroptosis induced by SCI plays a crucial role in worsening nerve injury and impeding regeneration. Excess iron ions elevate ROS production *via* the Fenton reaction, triggering neuronal apoptosis and disrupting the regenerative microenvironment [191, 192]. Geng *et al*. constructed a natural polyphenol-based bio-adaptive hydrogel based on the antioxidant and iron chelating properties of natural polyphenols (such as gallic acid and catechin) [193]. The hydrogel can remove excess iron ions from the injured area *via* chelation by polyphenol groups, inhibit the Fenton reaction and ROS generation, and reshape neural iron homeostasis. The anti-inflammatory and neuroprotective activities of natural polyphenols can further

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improve the SCI microenvironment, activate the endogenous nerve regeneration mechanism, promote the repair and reconnection of neural pathways across the injury site, facilitate the formation of functional synaptic connections, and ultimately promote motor function recovery in SCI rats.

5.3 Reconstruction of neural circuit

Reorganization of neural circuits that underlie the recovery of motor function after SCI. This functional reconstruction relies not only on structural integrity but also on signaling accuracy. Bio-adaptive hydrogels can guide axons across injury scars, grow directionally toward targets, and reconstruct nerve conduction pathways; they can also simulate the bionic environment of the spinal cord, induce the sprouting of axon collaterals of descending conductive tracts such as the CST, and establish functional synapses with interneurons distal to the injury, thereby building alternative neural circuits across the lesion area.

Remodeling of corticospinal circuits depends on neuronal activity. Studies have found that after injury, CST axons sprout and form new synapses with spinal interneurons. This process is maintained by NMDAR/CREB signaling and selects effective connections through activity-dependent competition [194]. Using strategies such as electrical stimulation to enhance the physiological activity of relay neurons can strengthen corticospinal connections and promote motor function recovery. Conductive hydrogels can achieve this electrophysiological regulation. Meanwhile, neural stem cell grafts can form spontaneously active synaptic networks at the injury site and functionally integrate with host neurons in a physiological pattern similar to that of a normal spinal cord (**Figure 9B**) [195]. Combining the two, that is, using conductive hydrogels loaded with NSCs, can synergistically promote the reconstruction of neural circuits and the recovery of motor function.

For example, the NGP hydrogel designed by Liu *et al*., loaded with NSCs and memantine, can not only significantly promote NSC neuronal differentiation and axonal growth but also effectively restore nerve conduction function. Experimental results showed that the hydrogel significantly increased myelin area, the number of new neurons, and axon area, while minimizing cyst cavity and glial scar areas, thereby promoting the reconstruction of neural circuits. After 6 weeks of implantation, the BBB

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score reached a maximum of 10 points. The electromyogram amplitude returned to a level closest to that of normal rats, jointly confirming its significant effect in promoting motor function recovery [175]. However, amplitude recovery only reflects the amplitude of motor nerve impulses transmitted to muscles at the electrophysiological level. The assessment dimension is too single and cannot assess fine motor control and sensory functions.

The axons of calcitonin gene-related peptide-positive ($CGRP^+$) neurons in the dorsal horn connect to secondary neurons in the corresponding spinal cord segments via synapses, transmitting primary sensory information to the spinothalamic tract and, ultimately, to the thalamus [196, 197]. The DA-grafted hyaluronic acid supramolecular hydrogel developed by Tan *et al*. can promote axon regeneration of primary afferent sensory neurons and reconstruct nerve conduction pathways. Owing to the self-healing property and excellent tissue adhesion ability of the hydrogel, it can adhere well to the spinal cord and induce the regeneration of a large number of CGRP neurons within the hydrogel, ultimately achieving excitatory synaptic connections between the hydrogel and the rostral boundary of the injury **(Figure 9C-E)** [184]. At the same time, the hydrogel improved the survival rate of V2a interneurons in the injured area. Such neurons serve as transfer hubs that can bridge ascending and descending neural pathways in a target-specific manner without the need for long-range axonal regeneration [198, 199]. This strategy of establishing heterogeneous neural connections *via* neuronal relays is not only effective in a rat SCI model with intact anatomy. It has also been verified in a beagle model that more closely resembles human injury characteristics, further enhancing its clinical translation potential.

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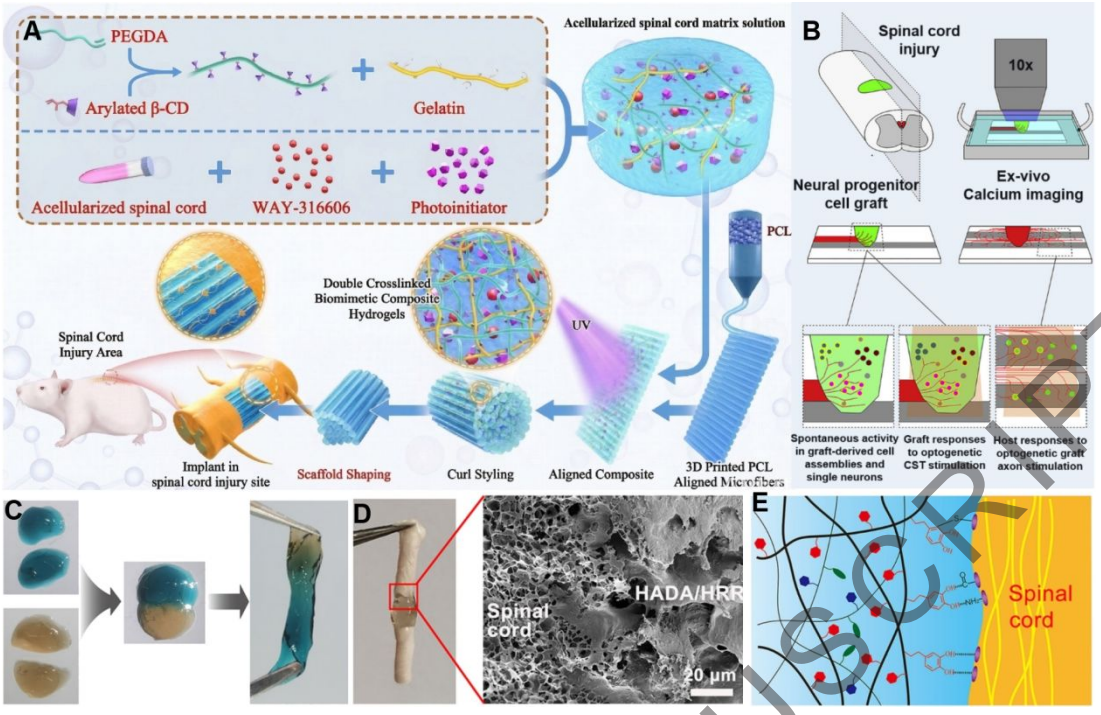


Figure 9. (A) Preparation of Aligned Fiber Hydrogels. Adapted from Ref. [51] with permission of Elsevier, © 2022. (B) Schematic diagram of a neural stem cell graft forming a synaptic network at the injury site. Reproduced from Ref. [195] with permission of Elsevier, © 2020. (C) Self-healing behavior of hydrogel. (D) Digital photos and SEM images at the junction of the hydrogel and spinal cord. (E) Schematic illustration of the interactions between the hydrogel and the native spinal cord. Reproduced from Ref. [184] with permission of American Association for the Advancement of Science, © 2024.

6. Application of bio-adaptive hydrogel in motor recovery and bladder function restoration

The neural network located in the lumbosacral spinal cord, commonly referred to as the central pattern generator (CPG), consists of overlapping, functionally shared interneuronal networks that participate concurrently in the regulation of rhythmic locomotion and micturition reflexes [200-202]. This network forms intersegmental integrative circuits with intrinsic spinal projection neurons; interneurons within these circuits can simultaneously innervate lower-limb motor neurons, parasympathetic preganglionic neurons of the bladder, and motoneurons controlling the external urethral sphincter (EUS), thereby providing a synaptic basis for shared somatic and autonomic control [203, 204]. At the same time, ascending and descending pathways that govern motor and urinary functions also substantially overlap: corticospinal, reticulospinal tracts, and descending pontine micturition pathways project *via* the spinal white matter

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to the lumbosacral cord, while afferent sensory input from the bladder and lower limbs is conveyed to the CNS through similar ascending routes [205-207]. After SCI, these three major neural networks undergo coordinated plastic changes. In the acute phase, loss of descending control combined with global suppression of network excitability results in flaccid paralysis and areflexic urinary retention [208]. In the chronic phase, disinhibition of the CPG, aberrant synaptic rearrangement of intrinsic spinal neurons, collateral sprouting of residual fibers, and central sensitization give rise to abnormal coupling of locomotor and micturition circuits and synchronous dysfunction, manifesting as lower-limb spasticity and bladder overactivity [204, 209]. This voiding control disorder is central to neurogenic lower urinary tract dysfunction (NLUTD), typically manifesting as two hallmark symptoms: detrusor overactivity (DO) during the storage phase and detrusor-sphincter dyssynergia (DSD) during the voiding phase. Long-term high-pressure urination not only causes urinary incontinence and repeated urinary tract infections but also gradually damages renal function, becoming an important contributing factor for the persistently high mortality rate in the later stage of SCI patients [210, 211]. Compared with restoring the ability to walk, improving bladder control is often the primary demand for SCI patients seeking to enhance their quality of life. The anatomical and functional connections between motor neurons and sympathetic neurons of the autonomic nervous system provide a potential structural basis for coordinating bladder control *via* repair of motor circuits.

Research shows that hydrogel can effectively improve bladder function by repairing the spinal cord microenvironment and promoting neural circuit reconstruction. CD/HRI-NT3 hydrogel can not only promote nerve regeneration and sensory-motor synapse reconstruction, but also reduce bladder wall thickening, muscle bundle disorder, and other pathological changes by inducing tyrosine hydroxylase-positive (TH^+) axon regeneration and maintaining the regular arrangement and density of detrusor cells, thereby significantly restoring bladder function [160]. Similarly, a DA-grafted hyaluronic acid supramolecular hydrogel can improve bladder morphology and function by promoting TH^+ axon regeneration, restoring bladder volume to near-normal levels, and reducing corresponding pathological damage[184]. The realization of these effects mainly relies on the ability of spinal TH^+ interneurons to coordinate the activity of the bladder and EUS during the micturition reflex that occurs after SCI [212]. Further studies have shown that hydrogels can also enhance bladder afferent signaling

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and reconstruct glutamatergic micturition control circuitry by promoting the regeneration of $MAP2^+$ interneuron synaptic connections and VGluT2 [213]. SCI often causes urinary dysfunction with bladder overdistension, swelling, thinning, and increased transparency. Hydrogel treatment significantly alleviates these issues and restores bladder function (**Figure 10A, B**). The lumbosacral spinal cord serves as a common hub for movement and urination, and its repair can support both functions simultaneously. The GEL-EXO hydrogel designed by Lin *et al*. achieves dual recovery of movement and bladder function by repairing the distal lumbosacral spinal cord, protecting neurons, and improving the local microenvironment (**Figure 10C, D**) [214]. However, the mechanism by which it repairs the distal region remains unclear.

In addition, combining electrical stimulation with a conductive hydrogel offers a new approach to simultaneously improve motor and bladder function. Normal micturition relies on precise coordination between the detrusor and sphincter muscles by the supraspinal loop, and SCI often disrupts this coordination, leading to neurogenic bladder dysfunction [215, 216]. Zhang *et al*. combined electroacupuncture with GL-C40 conductive hydrogel and found that the combined therapy could restore bladder sensory conduction pathways by upregulating neuroactive ligand-receptor interactions, enhancing GABAergic signaling, promoting synaptic plasticity remodeling, and promoting axonal regeneration. At the same time, this combined therapy maintains the integrity of spinal cord electrical signals by coordinating calcium ion channel function, accurately regulating detrusor contraction rhythm, and ultimately improving bladder filling-voiding coordination. Further histopathological evaluation showed that although the detrusor and sphincter muscles of SCI model rats exhibited extensive tissue fibrosis, overall repair was satisfactory (**Figure 10E**) [217].

The brain areas that control the pelvic floor muscles and lower-limb movements overlap in space, and pelvic floor activities can effectively activate cortical areas involved in lower-limb movements [218]. At the same time, numerous studies have shown that restoring lower limb motor function plays a key role in regulating NLUTD. Targeted neuromodulation can inhibit detrusor overactivity, thereby promoting recovery from NLUTD [219, 220]. Interventions targeting shared lumbosacral spinal circuits (such as epidural electrical stimulation) can concomitantly improve somatic motor function and bladder voiding [221]. The above studies have fully revealed the close anatomical and

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physiological relationship between the exercise control circuit and the urinary circuit, providing a theoretical basis for restoring bladder function by repairing the exercise pathway. Current research mainly focuses on the repair of single movement function, and there are relatively few studies on restoring bladder function. In the future, more attention should be paid to the dual repair of movement and urinary function, as well as to the coupling mechanism between the two.

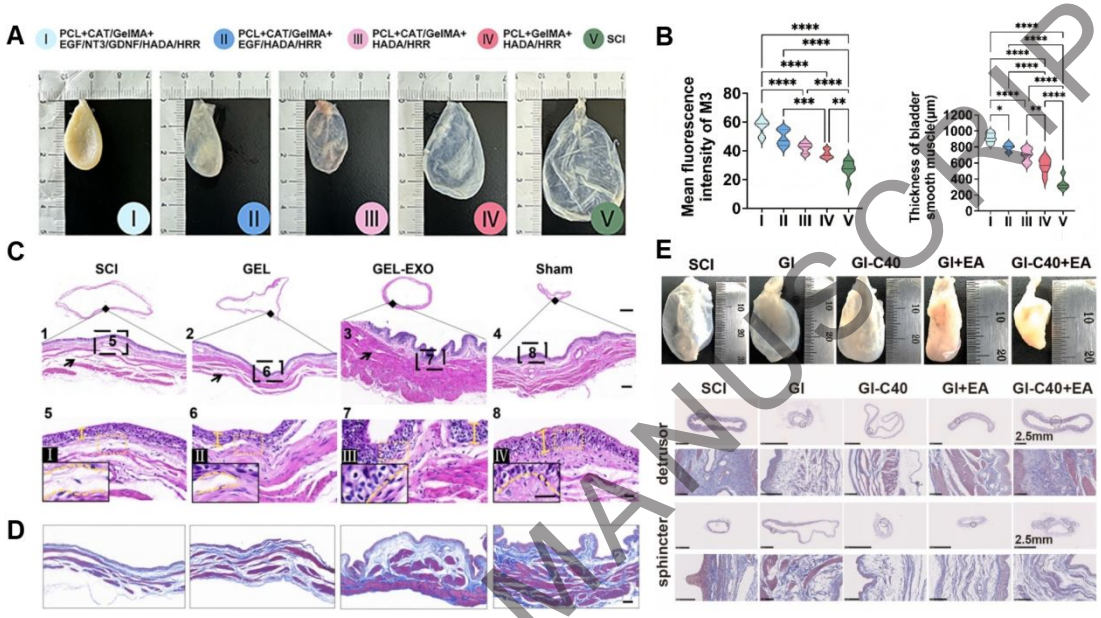


Figure 10. (A) Effect of hydrogel on pathological bladder changes. (B) Quantitative statistics of bladder muscle thickness and average fluorescence intensity of M3 in the bladder wall after repair. Reproduced from Ref. [213] with permission of Wiley-VCH, © 2026. (C, D) H&E and Masson staining images of the bladder after treatment. Reproduced from Ref. [214] with permission of Elsevier, © 2025. (E) Representative images of the bladder and its Masson staining images. Adapted from Ref. [217] with permission of Elsevier, © 2025.

7. Challenges and Limitations

The treatment strategy based on bio-adaptive hydrogels offers numerous significant advantages for SCI treatment. By constructing a three-dimensional scaffold that simulates the ECM, it provides physical support for the damaged spinal cord and Bridges the injured ends, guiding the orderly extension of axons. At the same time, it can regulate excessive inflammatory responses in the injury area, inhibit the formation of glial scars and oxidative stress-induced damage, and reshape a microenvironment conducive to nerve regeneration. By integrating functions such as conductivity, targeting, and responsive release, bio-adaptive hydrogels can synergistically promote

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neuronal survival, axon regeneration, and myelin sheath reconstruction, providing a highly promising strategy for the recovery of neurological function after SCI. In addition, injectable hydrogels offer minimally invasive implantation, in situ gelling, and good compatibility with host tissues, thereby adapting to the complex injury patterns of the spinal cord and reducing the risk of secondary injury. However, bio-adaptive hydrogels still face many challenges in practical applications.

7.1 Biological complexity

The post-SCI microenvironment is intricately dynamic and complex. Secondary damage mechanisms, including ferroptosis, oxidative stress, inflammatory cascades, and scar formation, impede nerve regeneration. SCI results in vascular rupture and disruption of the BSCB, leading to ischemia, hypoxia, edema, and inflammation, thereby exacerbating neuronal loss [8]. During the acute phase of injury, neutrophils and microglia rapidly infiltrate, eliminating necrotic tissue and debris, reducing the diffusion of injury-related model molecules, and preventing the expansion of secondary injury. Microglia secrete anti-inflammatory molecules, such as IL-4 and IL-10, and neurotrophic factors, such as Brain-Derived Neurotrophic Factor (BDNF) and NGF, promoting angiogenesis and the migration and differentiation of NSCs. Astrocytes form loose scars to physically isolate the damaged area, prevent the spread of inflammation, and secrete ECM components to provide a temporary scaffold for regeneration. During the chronic stage of SCI, the microenvironment tends to deteriorate. Pro-inflammatory molecules such as TNF-α and IL-1β, as well as ROS and proteases, are excessively released, causing excitotoxicity, lipid peroxidation, axon demyelination, and expanding the range of neuronal death. Excessive proliferation of astrocyte scars and the formation of dense scars severely impede axon extension. Abnormally activated microglia continuously produce inflammatory mediators, exacerbating neuronal apoptosis[222-224]. The optimal design of bio-adaptive hydrogels must facilitate multi-dimensional coordinated regulation. This includes modulating inflammatory responses, offering directional topological guidance to promote the orderly crossing of axons across the injury site, supporting vascular reconstruction, and fostering the formation of neural circuits and stable synaptic integration, ultimately restoring neural signal transduction. Currently, however, most bio-adaptive hydrogels exhibit only a single function or a limited range of functions. This limitation hampers the ability to address the

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aforementioned multiple requirements, thereby reducing neural regeneration efficiency and transient functional recovery.

7.2 Assessing complexity

There is still no unified, comprehensive evaluation system for SCI repair outcomes. The existing evaluation methods are difficult to use to comprehensively and accurately assess the repair value of bio-adaptive hydrogels. This problem restricts the in-depth development of related research. At present, *in vivo* assessment mainly focuses on motor function indicators (such as the rat BBB score, inclined plate test, and gait analysis). Still, the functional disorders of SCI patients are multidimensional. Besides the loss of motor function, they are often accompanied by sensory dysfunction, autonomic nerve dysfunction, such as neurogenic bladder, and complications such as pain. Focusing solely on motor function will lead to a one-sided assessment of the repair effects of bio-adaptive hydrogels. Some bio-adaptive hydrogels may promote motor nerve regeneration, but they cannot improve sensory or bladder function, thereby limiting their clinical translation potential. The currently widely used behavioral assessment is at the macro level, and the operator's subjective factors easily influence evaluation results. It lacks objective micro-level reading that reflects functional states, such as synaptic conduction efficiency and neural circuit reconstruction. It is difficult to elucidate the internal mechanisms by which bio-adaptive hydrogels regulate neural regeneration and functional recovery, nor can they accurately assess the stability and durability of the repair effect. Furthermore, existing evaluations of neurogenic bladder predominantly focus on macroscopic voiding behavior, lacking in-depth analysis of neural circuit remodeling associated with bladder function. Therefore, establishing a multi-dimensional, standardized *in vivo* evaluation system encompassing motor, sensory, and autonomic functions, as well as neural circuit remodeling, while simultaneously incorporating neurogenic bladder function and corresponding neural circuit reconstruction as core assessment endpoints, is a critical issue urgently requiring resolution to overcome current evaluation limitations and advance the clinical translation of SCI repair research.

7.3 Difficulties in technology transformation

Existing research primarily focuses on acute SCI models in small animals, such as mice and rats. However, most clinical SCI patients have chronic injuries, and the spinal cord

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structure and injury pathological process of large animal models are closer to those of humans. But there are relatively few related studies, which makes it difficult to directly translate the repair effects of bio-adaptive hydrogels into clinical practice. In addition, when bio-adaptive hydrogels move from the laboratory to clinical practice, they still face many challenges, such as preparation processes, large-scale production, and material stability. For instance, bio-adaptive hydrogels with directional topological structures can achieve precise regulation in small-scale laboratory preparations. However, when produced on a large scale, it is difficult to ensure uniformity of the topological structure, and production costs increase, making it unsuitable for clinical batch applications. The three-dimensional porous structure and bio-adaptive components of bio-adaptive hydrogels are sensitive to sterilization conditions, which can easily lead to performance degradation and loss of biological activity. Most bio-adaptive hydrogels have a short shelf life and are difficult to store and transport for long periods. Moreover, the preparation of bio-adaptive hydrogels requires the use of bio-adaptive substances, polymer materials, and complex preparation processes, resulting in costs much higher than those of conventional medical materials and limiting their clinical popularity. These problems collectively make it difficult for hydrogels with strong therapeutic effects in the laboratory to advance to clinical verification and industrialization.

In conclusion, the application of bio-adaptive hydrogels in SCI research still needs to address issues such as limited repair efficiency, incomplete functional evaluation systems, and difficulties in clinical translation. To address these issues, in the future, it is necessary to enhance the cross-integration of fields such as neuroscience, materials science, regenerative medicine, pharmacy, and AI, develop multifunctional bio-adaptive hydrogels, establish a more comprehensive functional evaluation system, and promote the research and application of individualized customized bio-adaptive hydrogels, providing innovative, safe, and efficient treatment options for SCI patients.

8. Conclusion and prospects

SCI is a serious neurological disorder with a relatively high global incidence rate, mainly caused by falls, car accidents, and violent trauma. The irreversible damage to neurological function it causes often leads to permanent motor dysfunction or even paralysis in patients. Bio-adaptive hydrogels, with their excellent biocompatibility, ease

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of operation, and three-dimensional porous structures that mimic natural extracellular matrices, have become the main focus of SCI repair research. Bio-adaptive hydrogels can not only serve as scaffolds to provide mechanical support to the injured area, but also prevent syringomyelia and excessive scar tissue proliferation. Moreover, they can actively regulate the microenvironment of injury and promote nerve regeneration [59, 225-227]. This review systematically summarizes the design strategies for bio-adaptive hydrogels with directional topological structures, electrical conductivity, stimulus responsiveness, and adhesion. It focuses on elucidating the role of these hydrogels in actively regulating repair after SCI and in inhibiting scar tissue formation. It particularly explores the potential mechanisms underlying their synergistic recovery of locomotion and bladder function. Looking to the future, designing multifunctional, individually tailored bio-adaptive hydrogels will be a promising research direction.

In previous studies, the development of bio-adaptive hydrogels has largely relied on empirical trial-and-error approaches, leading to prolonged development cycles, high costs, and difficulty in precisely capturing governing relationships in multi-component formulation screening, crosslinking structure regulation, and multi-performance co-optimization. These limitations hinder the ability to meet the customized requirements of complex application scenarios. The integration of AI and machine learning (ML) holds promise for rejuvenating the development and application of bio-adaptive hydrogels. Through data-driven modeling, rapid material discovery, accurate performance prediction, and formulation optimization, AI/ML can reduce design costs and enable precise customization, thereby markedly improving development efficiency and the controllability of material properties [228-230]. By leveraging extensive databases encompassing polymer types, crosslinking modalities, processing parameters, and environmental conditions, AI can construct predictive models for hydrogel swelling behavior, mechanical strength, degradation rate, adhesive properties, and drug release kinetics, enabling rapid assessment and multi-objective optimization of key performance metrics [228, 231, 232]. AI has already been applied to the selection of molecular building blocks for hydrogels, formulation optimization, and the design of novel bio-adaptive peptides [233, 234]. In bioprinting, AI can integrate rheological parameters with printing protocols to accurately predict hydrogel printability and scaffold fidelity, thereby supporting the efficient fabrication of scaffolds for bone, cartilage, and neural tissue engineering [235]. For stimuli-responsive hydrogels, ML

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can establish structure–function relationships between external cues (pH, temperature, redox state, etc.) and drug-release behavior, thereby enabling temporally controlled delivery from drug-loaded systems. AI is likewise capable of predicting hydrogels' physical responses, such as swelling and compression, guiding targeted synthesis and precise performance tuning [236, 237]. In the design of medical adhesives and conductive hydrogels, AI-enabled inverse design can optimize monomer composition and network architecture to yield multifunctional hydrogels with high underwater adhesion, favorable conductivity, and biocompatibility, suitable for wound closure, wearable sensors, and neural interfaces [233, 238]. Collectively, these studies demonstrate that AI-driven strategies can effectively shorten hydrogel development timelines, reduce experimental consumption, and shift hydrogel research from empirical practices toward precision and intelligence. Current foundational AI models applied to the research and development of hydrogels for SCI repair are primarily classified into two categories: classical ML models and deep generative models. Traditional ML models, including random forest (RF), support vector machine (SVM), and artificial neural network (ANN), are primarily used for inverse design, performance prediction, and optimization of formulation parameters in hydrogels [239-241].

In contrast, deep models such as HydrogelFinder and convolutional neural networks (CNNs) exhibit more diverse and expanded capabilities. Specifically, HydrogelFinder enables the directed design of self-assembling peptides and functionalized macromolecular structures tailored to the repair demands of SCI [242]. At the same time, CNN can intelligently identify microscopic structural features, surface morphologies, and fabrication defects in hydrogels, thereby facilitating not only material structural characterization and manufacturing quality control, but also the optimization of microfluidic channel design and 3D printing process parameters[231]. Future research should focus on developing standardized SCI-specific databases and achieving end-to-end integration of compositional parameters, physicochemical properties, and *in vitro* and *in vivo* therapeutic efficacy data. By leveraging mature AI models for high-throughput virtual screening and process optimization, the ultimate goal is to establish an "AI-assisted personalized hydrogel" research and development platform to promote efficient and precise SCI repair.

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The complex and harsh microenvironment after SCI, characterized by both temporal dynamics and spatial heterogeneity, is a core bottleneck hindering endogenous neural regeneration, the survival of exogenous stem cells, and functional repair. However, the current traditional intervention mode of a single target, single time course, and single dimension is difficult to adapt to the dynamic evolution of the pathological process in SCI. This is also a key scientific and clinical challenge that urgently needs to be overcome in the field of neural repair. Based on the temporal characteristics of the SCI pathological process. For instance, during the acute phase, this intelligent hydrogel can release anti-inflammatory drugs to inhibit secondary damage. Subsequently, in response to changes in the microenvironment after inflammation subsides, hydrogels automatically switch to promoting nerve regeneration, combining topological structures, such as directional fibers, to guide axon growth. In the later stage, hydrogels can respond to external electrical stimulation, promote the reconstruction of neural circuits and signal transduction, and achieve multidimensional functional recovery, including movement and autonomic nerve function. This temporally responsive hydrogel can precisely match the entire process requirements for damage repair, addressing the limitation that single-function hydrogels are difficult to adapt to complex, dynamic repair processes. It should be noted that patients with SCI injuries exhibit individual differences, including injury location, injury severity, age, and immune status. A single formula of bio-adaptive hydrogel is difficult to be suitable for all patients. Personalized customization is the core development direction for future precise SCI repair. The mechanical properties of the hydrogel can be regulated according to the severity of the injury. For mild injuries, a formula with low hardness and high biological activity can be used to promote nerve regeneration. For moderate-to-severe injuries, the hardness should be increased to provide mechanical support, and a scaffold should be used to maintain the spinal cord's shape. At the same time, drug combinations can be tailored to patients' specific pathological characteristics. For instance, anti-inflammatory components can be added for patients with strong inflammatory responses, and the immune compatibility of hydrogels can be optimized for those with weakened immune functions. Under the above-mentioned precise regulation, it is expected that individualized repair plans will be developed, thereby enhancing therapeutic effects and promoting clinical transformation.

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In addition, this review clearly proposed the exercise-bladder coupling hypothesis, which provides a new theoretical perspective for the multi-functional collaborative repair of SCI and is also a key direction for future research and development of hydrogels. In the future, efforts can be made to develop a bio-adaptive hydrogel that combines conductivity and physical anisotropy. This hydrogel can load active agents that promote CST regeneration and provides strong protection for the sacral parasympathetic neurons that innervate the bladder. By combining electrical stimulation with an anisotropic topological structure to provide directional guidance, and by simultaneously guiding the downward movement of axons and the autonomic nerve fibers related to urination across the injury area, electrical stimulation provides physical guidance. It simultaneously guides the downward movement of axons and autonomic nerve fibers involved in urination across the injury site, thereby reconstructing neural circuits and ultimately enabling coordinated recovery of motor and urinary functions.

Declaration of Competing Interests

The authors declare no competing financial interests.

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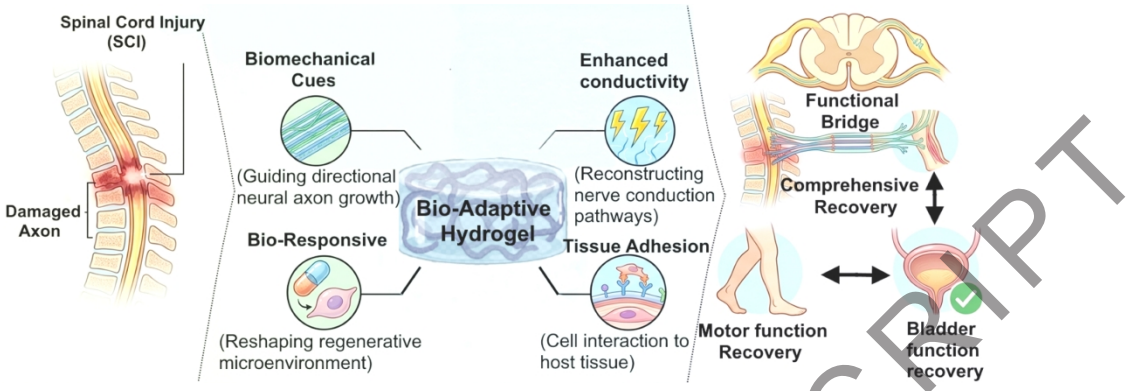
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